CP001824.1:1117181-1133702
Sphaerobacter thermophilus DSM 20745

B12-binding Radical_SAM Methyltransf_11 MTS Methyltransf_25 B12-binding Radical_SAM B12-binding Radical_SAM ? ? ParE_toxin PhdYeFM_antitox NTP_transf_2 ABC_membrane ABC_tran 2-Hacid_dh_C Beta-lactamase2 Na_H_antiporter TENA_THI-4

CP001774.1:1044-7574

Spirosoma linguale DSM 74

BLUF ? ? ? ParE_toxin ? Rep3_N ? NikA-like Relaxase ? PfkB

CP008939.1:40171-54532
Candidatus Paracaedimonas acanthamoebae

? ? DDE_Tnp_IS240 DEDD_Tnp_IS110 Transposase_20 HTH_Tnp_1 ? DDE_Tnp_IS240 ParE_toxin DUF6290 ? HTH_Tnp_1 Transposase_20 DEDD_Tnp_IS110 ? ? DUF3310

AP018223.1:1024859-1037851
Nostoc linckia NIES-25

Acetylase Orn_Arg_deC_N DAC Prenyltransf ParD_antitoxin DUF3143 DnaJ ParE_toxin RelB-like DnaJ DnaJ_C HSP70 Peptidase_M28 Peptidase_M28 ? ? DUF559

CADCTY010000587.1:899-893
uncultured Leptolyngbya sp.

? ParE_toxin

CVOA01000139.1:177-5306
Chlamydia trachomatis

? RHH_1 ParE_toxin ? ? BrnT_toxin BrnA_antitoxin Resolvase HTH_23 Glucosamine_iso Abhydrolase_3

GL622194.1:979652-993815
Actinomyces sp. oral taxon 180 str. F0310

FtsX Pps ABC_tran Acetyltransf_7 Na_H_Exchanger Na_H_Exchanger Protoglobin DNA_binding_1 ? PhdYeFM_antitox ParE_toxin Nitroreductase YoeB_toxin RelB Beta_elim_lyase Methyltransf_25 RF-1 UDG

BEHR01000009.1:71700-83957
bacterium HR15

? Lactamase_B RNase_J_b_CASP RMMBL RNase_J_C Clp_N Clp_N AAA AAA_lid_9 AAA_2 ClpB_D2-small ? ? ? RelB-like ParE_toxin ? ? ? ? ?

DNUJ01000444.1:1766-5241
Nitrospira sp.

? ParE_toxin ScsC_N DSBA NT5C TPRs ? ? TPRs PGAP1

DQWU01000172.1:2537-3341
Candidatus Bathyarchaeota archaeon

AAA DSHCT ParE_toxin

DASSAY010000031.1:124-3943
Balneolaceae bacterium

ParE_toxin PhdYeFM_antitox Aminotran_3 Acetyltransf_7 Ala_racemase_N

DASSKK010000053.1:14668-27213
Nocardioidaceae bacterium

DUF5305 ParE_toxin Peptidase_S26 ? Phosphodiest Transposase_29 PhdYeFM_antitox ParE_toxin HTH_23 Helicase_C DEAD HicB PIN Synaptojanin AbiEii

DASTLV010000184.1:10-2600
Nitrososphaera sp.

? ? ParE_toxin RelB_like_antitoxin RelB-like ParE_toxin RelB-like

DASTWU010000182.1:52-12232
Gemmatimonadaceae bacterium

? HTH_8 AAA_lid_14 NtrC-AAA Methyltransf_25 Acetylase PhdYeFM_antitox ParE_toxin PadR MacB_PCD FtsX MacB_PCD FtsX FR47 DUF4126 DUF2911 OGG ?

DASYVT010000060.1:1-8782
Candidatus Methylomirabilota bacterium

MFS_1 Leu_Phe_trans Fumerase_C Fumerase HMGL-like HCS_D2 LeuA_dimer ParE_toxin PhdYeFM_antitox CoA_transf_3 zf-ChsH2 OB_ChsH2_C Thiolase_N Thiolase_C_1 HTH_8 ABC_sub_bind

DASYYM010000973.1:236-9771
Puia sp.

? ORF6N Phage_integrase Phage_int_SAM_1 Phage_integrase Mg_chelatase Toprim_2 DNAG_N Zn_ribbon_DnaG HTH_3 RelB-like ParE_toxin ? ?

DASZBR010000005.1:86063-98211
Gammaproteobacteria bacterium

TPP_enzyme_N TPP_enzyme_M TPP_enzyme_C YecM Pyr_redox_2 HTH_1 LysR_substrate HTH_18 GyrI-like Beta-lactamase RelB-like ParE_toxin ? MFS_1 Dabb HTH_1 LysR_substrate Acetylase Acetylase Fic DUF4172

DASZFI010000197.1:3741-6146
Gemmataceae bacterium

Proton_antipo_N Oxidored_q2 Oxidored_q3 NADHdh SMC_ScpB ParE_toxin PhdYeFM_antitox NTP_transferase LbH_GLGC

DASZIA010000098.1:1-7723
Planctomycetaceae bacterium

? PQQ_2 PQQ_2 PhdYeFM_antitox ParE_toxin FloA ? Glyco_transf_4 Glycos_transf_1 MoCF_biosynth CinA_KH CinA

DASYML010000235.1:2419-11847
Pyrinomonadaceae bacterium

XdhC_CoxI XdhC_C Mpo1-like XdhC_CoxI ? QRPTase_N QRPTase_C tRNA-synt_1 Anticodon_1 Val_tRNA-synt_C Receiver ParE_toxin T2SSG

DATAHG010000705.1:1-3550
Thermoanaerobaculia bacterium

DEAD Helicase_C WH_Lhr DEAD_assoc ParE_toxin RelB-like dUTPase ?

DTUX01000200.1:3433-5579
Aquificales bacterium

N6_Mtase HsdM_N HepT-like Polbeta Methylase_S ParE_toxin RHH_1 Unstab_antitox SWI2_SNF2 REase

DQTY01000374.1:94-1676
Anaerolineae bacterium

Methyltransf_11 ParE_toxin PhdYeFM_antitox PhdYeFM_antitox

DYDX01000025.1:43671-55912
Rickettsia endosymbiont of Pyrocoelia pectoralis

YbeY Radical_SAM LIAS_N SHMT Abhydrolase_2 Glutaredoxin HhH-GPD EndIII_4Fe-2S PhdYeFM_antitox ParE_toxin DNA_gyraseA_C DNA_gyraseA_C DNA_topoisoIV ? ATP_bind_3 FtsH_ext AAA AAA_lid_3 Peptidase_M41 Fer2_3 Fer4_17

DATFHN010000029.1:133806-9768
Pseudonocardiaceae bacterium

Glycos_transf_4 Sua5_yciO_yrdC ParE_toxin PhdYeFM_antitox PrmC_N MTS PCRF RF-1 Ribosomal_L31 ? Rho_N Rho_RNA_bind ATP-synt_ab GHMP_kinases_N GHMP_kinases_C

DATEXI010000628.1:1-6394
Polyangia bacterium

? HAD_2 MMR_HSR1 YchF-GTPase_C PhdYeFM_antitox ParE_toxin HTH_8 Bps

DATFWS010000259.1:560-4222
Candidatus Binatia bacterium

? Rap1a SWIM PhdYeFM_antitox ParE_toxin Peptidase_S74 ABC_tran_CTD ?

DATFPH010000198.1:1-9160
Thermoanaerobaculia bacterium

tRNA-synt_2 N-OB Hexapep Hexapep ? Glycos_transf_2 RelB-like ParE_toxin Uma2 Fer2 Fer2_2 Ald_Xan_dh_C MoCoBD_1 MoCoBD_2 FAD_binding_5 CO_deh_flav_C MoxR-AAA ? PemK_toxin

DATEBB010000038.1:1-2436
Dongiaceae bacterium

ACCA PhdYeFM_antitox ParE_toxin SEC-C SecA_SW

DATDPG010000118.1:1-7957
Bdellovibrionota bacterium

TPR_CcmH_CycH Zn_ribbon_4 ? ParE_toxin AICARFT_IMPCHas MGS ORF6N ? ? MlaD ABC_tran MlaE

DATDOF010000804.1:310-2816
Chitinophagaceae bacterium

HTH_18 AraC_binding ParE_toxin RelB-like HTH_18

DATGLZ010000295.1:1-5113
Chloroflexota bacterium

BPD_transp_1 BPD_transp_1 PhdYeFM_antitox ParE_toxin Sulfatase Glyco_hydro_39 LacI Peripla_BP_3

DATHAT010000173.1:1-3147
Tepidisphaeraceae bacterium

HATPase_c_3 ? ParE_toxin ? ? Peptidase_M29

DATGVO010000021.1:1-3797
Planctomycetaceae bacterium

Receiver Receiver His-Kinase His-Kinase PAS PAS Receiver ? PhdYeFM_antitox ParE_toxin

DATGFC010000106.1:49934-62951
Candidatus Sulfotelmatobacter sp.

HIT S1 S1 S1 S1 S1 S1 Pyr_redox_dim Pyr_redox_2 HicA_toxin HicB_lk_antitox LplA-B_cat E1_dh ParE_toxin PhdYeFM_antitox Transketolase_C Transket_pyr ? Cytochrome_CBB3 Amidohydro_3 HTH_31 Cupin_2 Fasciclin

DAOSNS010000005.1:90606-103488
Anaerolineae bacterium

? FA_desaturase Sigma70_r2 Sigma70_r4_2 ? ? ? Radical_SAM SPASM ParE_toxin RelB-like ? Radical_SAM SPASM PqqD PqqD DUF2442 CN_hydrolase

DAONBX010000031.1:1399-14644
Aeromicrobium sp.

Melibiase_2 MFS_1 MFS_1 RibD_C Abhydrolase_1 SGNH Acyl_transf_3 THAP4_heme-bd PhdYeFM_antitox ParE_toxin IU_nuc_hydro UTRA GntR DeoC PfkB YchJ_M-like ? YibE_F

DAOLZS010000428.1:120-2929
Anaerolineales bacterium

dNK ? RelB-like ParE_toxin HD

DATIGG010000362.1:400-4082
Thermodesulfobacteriota bacterium

HicA_toxin HicB_lk_antitox ParE_toxin PhdYeFM_antitox PhdYeFM_antitox HTH_3 PIN RHH_1 PIN RHH_1

DATHQB010000142.1:1-6381
Candidatus Sulfotelmatobacter sp.

? ? Acyl-CoA_dh_N Acyl-CoA_dh_2 PhdYeFM_antitox ParE_toxin ? Sortilin-Vps10 Sortilin-Vps10 Peptidase_S49

DATHMO010000257.1:7263-15967
Bryobacteraceae bacterium

DNA_pol3_delta_C DNA_pol3_delta LptE Semialdhyde_dhC Semialdhyde_dh CDP-OH_P_transf PS_Dcarbxylase MotA_ExbB RelB-like ParE_toxin DUF3738 Glyoxalase_7 ? Glucosaminidase ? HTH_8 AAA_lid_14

DATHMV010000030.1:1-6434
Candidatus Aquilonibacter sp.

ECH_1 TraY Unstab_antitox ParE_toxin DUF2127 ? 3HCDH 3HCDH_N FitA-like_RHH ? ? Thiolase_C Thiolase_N

DATKAE010000023.1:3560-5008
Thermoanaerobaculia bacterium

STY-kinase PD40 PhdYeFM_antitox ParE_toxin TPRs

DATJXG010000114.1:1-6339
Terriglobales bacterium

Calcineurin ParE_toxin PhdYeFM_antitox Transketolase_C Transket_pyr Cytochrome_CBB3 Trans_reg_C Receiver His-Kinase His-Kinase HAMP ?

DATJPW010000022.1:564-10876
Desulfomonilaceae bacterium

Acetylase ? SdpI ? ParE_toxin PhdYeFM_antitox Methyltransf_25 ? AFOR_N AFOR_C Fer4_2 Fer4_11 N6_N4_Mtase Zn_ribbon_TFIIB Zn_ribbon_TFIIB ZnR Zn_ribbon_NUD

DATKCN010000221.1:206-6169
Thermomicrobiales bacterium

ParE_toxin PhdYeFM_antitox SnoaL Amidohydro_3 Lactamase_B_3 ? Uma2 3keto-disac_hyd

DATLAN010000286.1:7619-22781

Thermoguttaceae bacterium

? DUF6115 Hydrolase_4 ADH_N ADH_zinc_N UvrA_inter UvrA_DNA-bind AAA_29 ABC_tran ? RelB-like ParE_toxin PQQ_2 PQQ_2 Beta-prop_ACSF4 Beta-prop_ACSF4 Glyco_hydro_99 DUF5010 GH97_C Glyco_hydro_97

DATLAN010000585.1:1-15582

Thermoguttaceae bacterium

vWF_A Sigma70_r2 Sigma70_r4_2 Fer4_5 LdpA_Fe-S-bd Fer4_9 Fer4_2 SHD1 TPRs FMN_bind EFG_C EFG_IV EFG_III GTP_EFTU_D2 GTP_EFTU ParE_toxin PhdYeFM_antitox GFO_IDH_MocA_C2 GFO_IDH_MocA_C2 GFO_IDH_MocA ? Methyltransf_3 ATPase_2 DUF234 Acetyltransf_6

DATKZS010000071.1:67070-75126
bacterium

Receiver Receiver ? dCache_1 His-Kinase His-Kinase Receiver Peripla_BP_4 IU_nuc_hydro ParE_toxin PhdYeFM_antitox ?

DATKUA010000221.1:392-2310
Acidobacteriota bacterium

PhdYeFM_antitox ParE_toxin Peptidase_M24 Creatinase_N

DATLTK010000869.1:1-2775
Chloroflexota bacterium

? ParE_toxin RelB-like Gp53-like_C SH3_3

DATKMD010000077.1:12437-26771
Thermoanaerobaculaceae bacterium

LEA_2 Aminotran_5 Glycos_transf_1 Epimerase_2 Glycos_transf_N LpxK PfkB CTP_transf_like Phage_TAC_11 ParE_toxin PhdYeFM_antitox Peptidase_M54 Receiver NtrC-AAA AAA_lid_14 HTH_8 HAMP His-Kinase His-Kinase NrfD Fer4_7 Fer4_11 GCV_H

DATKJU010000297.1:106-4535
Thermodesulfobacteriota bacterium

Fer2_3 Fer2_2 ParE_toxin PhdYeFM_antitox Guanylate_cyc TPRs TPRs LemA

DASZZM010000078.1:446-9319
Thermoanaerobaculia bacterium

HTH_17 ? ? ParE_toxin ParE_toxin MacB_PCD FtsX MacB_PCD FtsX Unstab_antitox ParE_toxin

DASZPW010000203.1:299-3527
Chitinophagaceae bacterium

RelB-like ParE_toxin Peptidase_S37 Pyr_redox_2

DASZLR010000113.1:489-8142
Pseudonocardiaceae bacterium

PhdYeFM_antitox ParE_toxin PhdYeFM_antitox His-Kinase His-Kinase Trans_reg_C Receiver Ppx-GppA ? AP_endonuc_2 Pro_dh

DATMDN010001211.1:89-2627

Polyangiaceae bacterium

ParE_toxin PhdYeFM_antitox Sigma70_r2 Sigma70_r4_2 ? ?

DATFFW010000505.1:1-744
Candidatus Paceibacterota bacterium

PhdYeFM_antitox ParE_toxin

DATNNE010000147.1:1-3892

Terriglobales bacterium

Thiolase_C Thiolase_N BrnT_toxin CopG_antitoxin 3HCDH 3HCDH_N ParE_toxin TraY Unstab_antitox ? ECH_1

DATNHJ010000131.1:157-9605
Candidatus Acidoferrum sp.

Dehyd-heme_bind Peptidase_S10 Glyco_hydro_92N Glyco_hydro_92 RelB-like ParE_toxin BcrAD_BadFG GKRP_SIS_N GKRP-like_C ? PIN DivIC RRF AA_kinase ?

DATNHO010000194.1:25491-33668
Terriglobales bacterium

Ribosomal_S7 Ribosom_S12_S23 ? NusB DMRL_synthase ECH_1 TraY ParE_toxin 3HCDH 3HCDH_N Thiolase_C

DATNEW010000020.1:53208-66841
Gemmatimonadaceae bacterium

? HhH-GPD DUF6496 PAP2 BTAD GerE ? ParE_toxin PhdYeFM_antitox ? ? MutS_V PAP2 FUSC-like FUSC_2 HlyD_D23

DASPRL010000041.1:1-6582

Terriglobales bacterium

Pyr_redox_dim ? HicA_toxin LplA-B_cat E1_dh HTH_31 ? ParE_toxin PhdYeFM_antitox Transketolase_C Transket_pyr Cytochrome_CBB3 ?

DASQFH010000286.1:1-6067
Vicinamibacteria bacterium

GGDEF ParE_toxin PhdYeFM_antitox GatB_N Asparaginase_C Asparaginase GatD_N PemK_toxin RHH_1 ?

DASRBR010000691.1:1-7214
Verrucomicrobiia bacterium

UDPGP Transp_cyt_pur ParE_toxin ? Amidohydro_1 Aldedh 23S_rRNA_IVP DHO_dh

WBVI01000054.1:1-3901
Caedimonadaceae bacterium

DUF6290 ParE_toxin P-loop_TraG TraC_F_IV ? TrbI

LJUD01000249.1:1-5369

bacterium SM23_31

? ? ParE_toxin DUF1375 Peptidase_M48_N Peptidase_M48 DUF72 Peptidase_M20

JACDBA010000064.1:483-3903
Pseudonocardiales bacterium

ParE_toxin PhdYeFM_antitox ? GmrSD_N SLFN_AlbA_2

JACCWM010000100.1:1-14782
Lujinxingiaceae bacterium

HATPase_c_3 ? ? Transp_cyt_pur PhdYeFM_antitox ParE_toxin ? Amidohydro_1 Aldedh Aminotran_3 PseudoU_synth_2 S4 AbiEii Rv0078B

JACDGZ010000537.1:1-2243
Nitrospirales bacterium

Trypsin BPD_transp_2 ParE_toxin RelB-like

JACONN010000005.1:776667-794328
Hormoscilla sp. GM7CHS1pb

? ? Uma2 Uma2 ? DUF2281 PhdYeFM_antitox ParE_toxin PhdYeFM_antitox ? Uma2 ? Acetylase 5_nucleotid_C Calcineurin

JACMLH010000045.1:1-8384

Cytophagaceae bacterium

PemK_toxin TPRs TonB_C FGS Peptidase_S9 PD40 PhdYeFM_antitox ParE_toxin DUF3858 Transglutaminase DUF3857 Transglutaminase

JACMIR010000868.1:1-3776
Fibrella sp.

OEP HlyD_D23 dCMP_cyt_deam_1 ParE_toxin RelB-like RNB S1

JACNKN010000066.1:73214-92259
Candidatus Neomarinimicrobiota bacterium

Bgal_small_N LacZ_4 Glyco_hydro_2_C Glyco_hydro_2 Glyco_hydro_2_N Aminotran_3 APH Amino_oxidase Amino_oxidase Peptidase_M24 AMP_N ? DUF6290 ParE_toxin PAP2 Orn_Arg_deC_N Glyco_hydro_18 Ig_J_second Por_Secre_tail NamZ_N NamZ_C Peptidase_S13 DUF559 Lactamase_B_2

JACNIG010000176.1:978-11741
Candidatus Desulfatibia vada

Phage_integrase ? DUF4143 AAA_14 ? ? ? RHH_1 ParE_toxin LysM SLT Fer4 Fer4_7 Fer4_10 Fer4_5 Fer4_5 DUF362 Methyltransf_2 Dimerisation ?

JACVRQ010000519.1:1-3540
Coleofasciculus sp. Co-bin14

Glycos_transf_1 FTR1 ParE_toxin Uma2

JACJOK010000032.1:108533-118595
Trichocoleus sp. FACHB-90

? ? His_Phos_1 AstE_AspA_cat KH_2 MMR_HSR1 Lyase_1 ADSL_C Uma2 ParE_toxin ? ChlI Mg_chelatase Mg_chelatase_C HIT Photo_RC

JACJQI010000003.1:120470-141334
Aulosira sp. FACHB-113

TPS ? DuOB DUF488 DUF488 PIN bMG10 ParE_toxin ? Big Big bMG3 MG2 bMG5 bMG6 A2M_BRD A2M bMG10 BiPBP_C Transpeptidase Transgly ? Host_attach Receiver Trans_reg_C Hpt Receiver Cupin_2

JACZJZ010000512.1:1-2185
Deltaproteobacteria bacterium

Asp_decarbox ParE_toxin PhdYeFM_antitox Amidohydro_1

JACSRX010000035.1:70973-82313
Chthoniobacterales bacterium

? ? ? ? ? ? RelB-like ParE_toxin ? ? ? ? ? ? ?

SUUA01000002.1:403450-434066
Actinomyces succiniciruminis

DUF6880 ? SLFN_AlbA_2 HATPase_c_4 HTH_5 DUF3800 UvrD_C UvrD_C UvrD-helicase MZB Helicase_C DEAD PhdYeFM_antitox ParE_toxin Peptidase_M78 RelB MethyltransfD12 ? Eco57I MmeI_Mtase Helicase_C SNF2-rel_dom HTH_17

JADGCT010000232.1:1-3804
Desulfamplus sp.

? PKD K319L-like_PKD Big RelB-like ParE_toxin Bac_export_2

JADFYA010000125.1:614-8888
Deltaproteobacteria bacterium

Patatin ? ? AAA_16 His-Kinase His-Kinase PAS ParE_toxin PhdYeFM_antitox DUF433 AsnC_trans_reg2 DUF433 His-Kinase His-Kinase

JACOUI010000223.1:529-13587
Planctomycetia bacterium

DUF1501 N_methyl SBP_bac_10 ? PTP-NADK PALP PhdYeFM_antitox ParE_toxin ? YbjQ_1 HD GFO_IDH_MocA_C2 GFO_IDH_MocA TAT_signal DUF6807 ? Orn_Arg_deC_N

JACOTU010000879.1:1-2640
Candidatus Poribacteria bacterium

Thymidylate_kin DNA_pol3_delta2 DNA_pol3_delta2 DNA_pol3_delta2 DNApol3-delta_C PhdYeFM_antitox ParE_toxin tRNA-synt_1g

JACOXK010000068.1:28-10828
Deltaproteobacteria bacterium

GIY-YIG ? Sigma70_r4_2 Sigma70_r2 ACR_tran ACR_tran HlyD_D23 HlyD_D23 HMBD OEP OEP PhdYeFM_antitox ParE_toxin ? Haem_bd Cu-oxidase_2 Cu-oxidase Cu-oxidase_3

JACPFC010000066.1:16559-37089
Deltaproteobacteria bacterium

DUF2283 AAA_14 DUF4143 ? ? ? ? DUF6290 ParE_toxin DUF4143 AAA_14 NAGPA ATP-synt_ab Rho_RNA_bind Rho_N Lyase_1 ASL_C2 Orn_Arg_deC_N DAP_epimerase DAP_epimerase DHDPS

JACPFL010000004.1:11857-23557
Deltaproteobacteria bacterium

LigD_Prim-Pol DUF72 GSDH UbiD_N UbiD UbiD_C UbiD_N UbiD UbiD_C NMT1 Amidohydro_3 ParE_toxin PhdYeFM_antitox MFS_1 ABC_tran ABC_tran ?

JACPGD010000365.1:4676-9648
Candidatus Hydrogenedentota bacterium

Cep192_D4 ParE_toxin RelB-like OmpA TPR_NPHP3 TPRs Exonuc_VII_L

JACPJJ010000007.1:58352-74074
Planctomycetota bacterium

Sigma70_r4 Sigma70_r3 Sigma70_r2 Sigma70_r1_2 Hydrolase E1-E2_ATPase HMA Mur_ligase_C Mur_ligase_M MurD-like_N Cpn60_TCP1 HSP20 LON_substr_bdg AAA Lon_lid Lon_C RelB-like ParE_toxin ? Hexapep Hexapep Hexapep ? Transposase_20 DEDD_Tnp_IS110

JACPPF010000034.1:16369-35069
Planctomycetota bacterium

GFO_IDH_MocA GFO_IDH_MocA_C3 Bact_transglu_N Transglutaminase 3keto-disac_hyd TPRs ? FGGY_N FGGY_C PhdYeFM_antitox ParE_toxin ? DUF4058 Secretin TPRs ? TRAM PIN HD_assoc HD HIT

JACPPF010000114.1:91950-113027
Planctomycetota bacterium

? DUF4175 DUF4175 SQHop_cyclase_N SQHop_cyclase_C Peptidase_S41 PDZ_Tricorn Tricorn_C1 Beta-prop_THOC3 PD40 PD40 ? Uma2 Aldedh ParE_toxin RelB-like DUF4058 Methyltransf_31 AMP-binding DUF6036 ? PQQ_2 PQQ_2

JACPSE010000154.1:210-10653
Deltaproteobacteria bacterium

DUF2442 DUF4160 IspD CarD_TRCF_RID CarD_C Acetylase Fer4 Int_N Phage_integrase ParE_toxin PhdYeFM_antitox HTH_17 ? Zn_ribbon_DnaG HTH_17 ? HTH_17

JACPTQ010000044.1:97430-115332
Deltaproteobacteria bacterium

HTH_3 DUF4011 AAA_11 AAA_11 AAA_12 MTES_1575 REase SWI2_SNF2 T1R_D3-like T1RH-like_C DUF3644 Methylase_S HsdM_N N6_Mtase HTH_31 ParE_toxin PhdYeFM_antitox HTH_11 WYL WCX FHA Prok-RING_1 ? ? ?

JACPVI010000057.1:37478-59540
Myxococcales bacterium

DUF1638 GAF_2 His-Kinase His-Kinase OmpA CarboxypepD_reg CarboxypepD_reg ? Transglutaminase FHA NtrC-AAA AAA_lid_14 HTH_8 DDE_Tnp_1_4 DDE_Tnp_1_4 PhdYeFM_antitox ParE_toxin GxGYxYP_C GxGYxYP_N_3rd GxGYxYP_N_2nd GxGYxYP_N AAA_16 STY-kinase ? OPT STY-kinase DUF3060 DUF3060 ADH_N ADH_zinc_N

JACQED010000169.1:1696-8242
Chloroflexi bacterium

? HicA_toxin ATP-synt_DE ATP-synt_DE_N MreB_Mbl Put_DNA-bind_N CoA_binding HD_5 GAF_2 DUF1967 MMR_HSR1 GTP1_OBG ParE_toxin ? CTP_transf_like ?

JACQTE010000109.1:1-11566
Acidobacteria bacterium

PALP PRAI IGPS Sel1 Sel1 Sel1 Sel1 Sel1 Sel1 Sel1 Sel1 ParE_toxin PhdYeFM_antitox Lactamase_B_3 Acyl-CoA_dh_N Acyl-CoA_dh_M Acyl-CoA_dh_1 ? TIG CarboxypepD_reg Peptidase_M10 HhH-GPD STY-kinase

JACRCX010000047.1:114898-127017
Deltaproteobacteria bacterium

? ? ? MIF Amidohydro_3 Amidohydro_3 ? ? ParE_toxin PhdYeFM_antitox Aminotran_5 ACT PGDH_inter 2-Hacid_dh_C Radical_SAM

JACRDC010000044.1:25119-37943
Desulfarculus sp.

ParB_N PAPS_reduct 2-thiour_desulf HTH_ParB ParB_N Zn_ribbon_TFIIB ? GpA_ATPase GpA_nuclease PhdYeFM_antitox ParE_toxin ? Phage_portal_2 Phage_capsid_4 Phage_cement_2 Minor_tail_Z Phage_attach ?

JACRCO010000065.1:749-11938
Deltaproteobacteria bacterium

AbiEi_4 TfoX_N AHSA1 Aldo_ket_red Pro_dh PutA_N ORF6N PhdYeFM_antitox ParE_toxin DUF748 DUF748 Vgb_lyase Vgb_lyase DLH ? Glyoxalase_6 adh_short_C2 Isochorismatase

JACRIH010000248.1:12524-27712
Planctomycetales bacterium

? DinB_2 ? HMGL-like HCS_D2 LeuA_dimer PMT Synaptojanin Phage_int_SAM_4 Phage_integrase ParE_toxin PhdYeFM_antitox TubC_N Zn_ribbon_DnaG DUF3987 HTH_17 HTH_17 ? ?

JADJOF010000005.1:230988-242291
Bdellovibrionales bacterium

P-loop_TraG CagE_TrbE_VirB ? ? ? ? ? TrbI ParE_toxin RH3_dom ? HTH_17 Phage_integrase ? TrwB_AAD_bind ?

JADJPM010000001.1:36138-50772
Verrucomicrobiaceae bacterium

Cytochrom_C Epimerase Pur_ac_phosph_N ? OTCace_N OTCace ? Aminotran_3 ParE_toxin ? Receiver GerE His-Kinase His-Kinase PepSY_TM Glyco_hydro_26 TSP3_bac CBM_4_9 AA_kinase DciA

JADJTY010000001.1:44953-62092
Chitinophagaceae bacterium

Methyltransf_4 SDH_beta SDH_alpha DPPIV_N Peptidase_S9 Glyco_hyd_65N_2 Glyco_hydro_95_cat Glyco_hydro_95_C ? N-OB tRNA-synt_2 ? ParE_toxin ATP-synt_ab Rho_RNA_bind Rho_N ? cobW Rhodanese ? AIRS_C AIRS ? ?

JADJYZ010000006.1:195671-210610
Deltaproteobacteria bacterium

? ? WCX WYL HTH_11 ? ? DUF2237 Alpha-amylase Malt_amylase_C ParE_toxin ? Peptidase_S51 ACP_syn_III ACP_syn_III_C Chromate_transp Chromate_transp Cytochrom_C HYDIN_VesB_CFA65-like_Ig PEGA PEGA

JADHXM010000005.1:32186-40806
Desulfobacterales bacterium

Radical_SAM BATS ThiG ThiF ketoacyl-synt Ketoacyl-synt_C Receiver STY-kinase Usp ParE_toxin Antitox_RHH

JAEUWE010000360.1:18-1941
Bdellovibrionaceae bacterium

TrbI ParE_toxin RHH_10 ? HTH_17

JAEUUC010000007.1:27511-38650
Cyclobacteriaceae bacterium

? ? ? DUF5684 Lycopene_cyc Lycopene_cyc Cyt_bd_oxida_II Polyketide_cyc2 ParE_toxin RelB-like DNA_photolyase FAD_binding_7 DUF6435 NUDIX SQS_PSY Amino_oxidase Sigma70_r2 Sigma70_r4_2

JAEUQV010000086.1:1-15525
Candidatus Saccharibacteria bacterium

LPG_synthase_TM UbiA COX4_pro COX3 COX1 COX2_TM COX2 COX_ARM ParE_toxin DUF6290 FAD_binding_4 FAD-oxidase_C Pyr_redox_2 PaaX_M PDDEXK_1 UvrD_C UvrD_C UvrD-helicase MIP CDCA

VGFC01000178.1:308-14275
Armatimonadetes bacterium

DLH Peptidase_S9 PhdYeFM_antitox ParE_toxin tRNA-synt_1b AAA_3 AAA_lid_2 Transpeptidase Transgly SGSH_C Sulfatase He_PIG Rubrerythrin RUBY_RBDX Desulfoferrod_N Desulfoferrodox

VGFC01000216.1:1415-12541
Armatimonadetes bacterium

? Sigma70_r2 Sigma70_r4_2 CarboxypepD_reg ? ? ABC_tran ? Mrr_cat Helicase_C SNF2-rel_dom ParE_toxin HTH_17

JAFGJE010000237.1:14906-26775
Phycisphaerae bacterium

Pterin_bind DAC ? GTP_cyclohydro2 DHBP_synthase ? Memo PhdYeFM_antitox ParE_toxin FitA-like_RHH PIN Glyco_hydro_2_C Glyco_hydro_2 ? ? ? LacI Peripla_BP_3

JAFGRW010000100.1:488-11522
Spirochaetes bacterium

? Bax1-I ? RelB-like HTH_19 ParE_toxin RelB-like MCPsignal HAMP sCache_2 ? Big Big MFS_1 Bact_hydrolase AAT TonB_C

JAFGWX010000060.1:10052-17062
Methylothermaceae bacterium

PALP ? DUF1318 TRAM tRNA_U5-meth_tr YkuD UPF0114 RelB-like ParE_toxin GIIM

JAFITR010000011.1:11733-18088
Simkania negevensis

GDH_HM3 Bac_GDH_CD Bac_GDH_CD ELFV_dehydrog Hydrolase E1-E2_ATPase VIT1 ? ParE_toxin GDC-P GDC-P GcvP2_C Aconitase Aconitase_C LicD ? Glyco_transf_9 TLC

JAFKGL010000016.1:39943-53598
Candidatus Paracaedimonas acanthamoebae

TrbI_Ftype Peptidase_S26 ? ? TraU TrbC_Ftype F_T4SS_TraN ParE_toxin DUF6290 TraF TraH TraG_N ? TrwB_AAD_bind ?

JAGBKB010000073.1:24822-30762
Lachnospiraceae bacterium

Aldo_ket_red Lactamase_B_3 Flavodoxin_4 NUDIX TspO_MBR FMN_red Calcineurin DUF6290 ParE_toxin

JAFOEX010000001.1:260-3676
Spirochaetaceae bacterium

DUF6290 ParE_toxin Replicase PriCT_1 HTH_23 Mob_Pre DUF6290 ParE_toxin

JAFOLH010000003.1:185825-200473
Renibacterium sp.

Aldedh Sugar_tr Cupin_2 HTH_18 LssY_C ? SAICAR_synt GARS_N GARS_A GARS_C ParE_toxin PhdYeFM_antitox Asparaginase_II SCP_3 TctB Pribosyltran GATase_6 AIRS_C AIRS AbiEi_4

JAFSUP010000038.1:1-4089
Synergistaceae bacterium

GATase_3 N_methyl N_methyl ? ParE_toxin DNA_pol3_delta_C SprA-related

JAFVEW010000033.1:1-15759
Synergistaceae bacterium

Peptidase_S55 PCRF RF-1 Mem_trans Mem_trans SprA-related N_methyl DNA_pol3_delta_C ParE_toxin ? N_methyl N_methyl N_methyl POR_N PFOR_II POR EKR Fer4_7 Fer4_16 TPP_enzyme_C AMP-binding_C AMP-binding AMP-binding_C AMP-binding

JAANWY010000387.1:710-7069
Anaerolineales bacterium

APH ParE_toxin PhdYeFM_antitox UPF0029 DUF1949 TPP_enzyme_C Fer4_7 EKR POR PFOR_II POR_N

JAFDXN010000012.1:135234-143161
Bacteroidetes bacterium

? Por_Secre_tail DUF6089 NAD_kinase NAD_kinase_C Beta-prop_EML_2 Cytochrom_D1 PSCyt1 FolB FGAR-AT_linker AIRS AIRS_C AIRS AIRS_C ParE_toxin RelB-like ?

JAGVYZ010000020.1:11020-22739
Candidatus Woesearchaeota archaeon

Fic KilA-N ? ? Yip1 ? DUF7557 ParE_toxin tRNA_SAD tRNA-synt_2b HGTP_anticodon ? ? FmrO ? Flavin_Reduct S-AdoMet_synt_N S-AdoMet_synt_M S-AdoMet_synt_C

JABJOR010000579.1:1-3919
Rhodospirillales bacterium

MTTB MTTB HEPN_AbiU2 PhdYeFM_antitox ParE_toxin

JAHJDR010000326.1:1736-3509
bacterium

? AbiEi_2 ParE_toxin DUF6290 Pribosyltran Peptidase_M23

JAHJOU010000072.1:3097-7481
Bacteroidetes bacterium

CoA_transf_3 ParE_toxin RelB-like HMGL-like Iso_dh POR TPP_enzyme_C POR_N PFOR_II Fer4_7 Fer4_16

JAGWZK010000018.1:29535-42374
Planctomycetota bacterium

? MethyltransfD12 N_methyl SecA_SW SecA_SW P-loop_SecA SecA_DEAD SecA_DEAD ? TSCPD PhdYeFM_antitox ParE_toxin ? ? Calcineurin ? DUF4236 ? ?

JAGXBH010000050.1:5161-12474
Acidobacteria bacterium

DapB_C DapB_N Acyl-CoA_dh_N Acyl-CoA_dh_M Acyl-CoA_dh_1 MST Glyoxalase RibD_C HSP20 ? ParE_toxin PhdYeFM_antitox ?

JAFASY010000188.1:24867-30479

Chroococcidiopsidaceae cyanobacterium CP_BM_ER_R8_30

? ? XisI XisH Uma2 ParE_toxin ? ParE_toxin ? MazE_antitoxin ? ? ?

JAFAZW010000039.1:26327-46806
Alphaproteobacteria bacterium

Gal_mutarotas_2 Glyco_hydro_31_2nd Glyco_hydro_31_3rd DUF5110 Cu-oxidase_3 Cu-oxidase_2 BiPBP_C Transpeptidase Transgly bMG10 A2M A2M_BRD MG2 bMG3 ? ? ? ParE_toxin dnstrm_HI1420 ? Ppx-GppA FtsJ FMN_red Rick_17kDa_Anti Ribosomal_S13

JAFDJU010000469.1:1096-2526
Deltaproteobacteria bacterium

ParE_toxin PhdYeFM_antitox DUF2161

JAHWLB010000015.1:1-8973
Acidobacteria bacterium

? HTH_3 Toprim_2 DNAG_N Zn_ribbon_DnaG ? Phage_integrase Phage_int_SAM_4 ParE_toxin RelB-like ? ? His-Kinase DUF7713 DUF7686 ?

JAHHHX010000006.1:168939-181542

Plectolyngbya sp. WJT66-NPBG17

BPD_transp_2 BPD_transp_2 ABC_tran ABC_tran TPRs TPRs TPRs DUF2252 Dzip-like_N SRP_SPB SRP54 SRP54_N ParE_toxin DUF2281 Ribosomal_S16 KH_KhpA-B PhoH ? Glyco_hydro_1 ?

JAHHHL010000059.1:8557-23334
Lyngbya sp. HA4199-MV5

Peptidase_S9 PD40 ? FGase HemY_N TPRs TPRs TPR_MalT TPRs TPRs TPRs RNaseH-CATRA+Caspase ? ParE_toxin UPF0150 ParE_toxin LRRs LRRs LRRs LRRs LRRs REase RuBisCO_large_N RuBisCO_large RbcX RuBisCO_small Lipase_GDSL_2 AAA RuBisCO_activase_AAA_helical RuBisCO_small

JAHBUR010000012.1:73309-89451
Bacteroidetes bacterium

Abhydrolase_1 Calcineurin Na_Pi_cotrans Na_Pi_cotrans MS_channel_1st MS_channel_2nd MS_channel_3rd CitMHS CitMHS LapA_dom Radical_SAM CofH_C ParE_toxin PhdYeFM_antitox Radical_SAM CofH_C VitK2_biosynth ? ? Pan3_PK AAA_23 ORF6N

JAHBTI010000010.1:54923-69893
Bacteroidetes bacterium

VitK2_biosynth PNPase DUF1015 2-Hacid_dh_C GTP_EFTU GTP_EFTU_D2 EFG_III EFG_C LepA_C HTH_26 HKD ParE_toxin ? ? QueG_DUF1730 Fer4_16 PHA_synth_III_E Abhydrolase_1 POTRA POTRA ShlB Por_Secre_tail

JAIEMP010000013.1:37418-48525
Chitinophagaceae bacterium

Acyl-CoA_dh_N Acyl-CoA_dh_M Acyl-CoA_dh_1 Lipase_GDSL_2 OMPdecase DUF5615 DUF433 Aminotran_5 KYNU_C GDPD ParE_toxin RelB-like FMN_bind_2 Transgly_assoc FeS_assembly_P RsgA_GTPase ParA CarbopepD_reg_2 23S_rRNA_IVP 3HCDH 3HCDH_N

JAIOHW010000998.1:1-2128
bacterium

ParE_toxin PhdYeFM_antitox MIP Apiosidase Collagen_bind_2

JAIQGH010000017.1:4252-18279
Terriglobia bacterium

UnbV_ASPIC FG-GAP_3 FG-GAP_3 FG-GAP_3 FG-GAP_3 FG-GAP_3 FG-GAP_3 HZS_alpha HTH_Crp_2 cNMPBD Citrate_synt tRNA_edit DUF2461 RelB-like ParE_toxin Sigma70_r2 Sigma70_r4_2 ? DUF3106 Methyltransf_25 Trypsin PDZ_2 ? DUF3303

JAIQGH010000149.1:5528-7893
Terriglobia bacterium

? ? UbiD_N UbiD UbiD_C PhdYeFM_antitox ParE_toxin ? DDE_Tnp_IS66_C ?

JAIQGA010000082.1:1-11849
Terriglobia bacterium

DDE_Tnp_1 ? ParE_toxin PhdYeFM_antitox Rhodanese Rhodanese TPRs TPRs TPRs TPRs ? TMP-TENI Lyase_1 ADSL_C Lactamase_B_2

JAIQFW010000066.1:16406-22264
Terriglobia bacterium

? ? ? 2-oxoacid_dh E3_binding Biotin_lipoyl Cytochrome_CBB3 Transketolase_C Transket_pyr PhdYeFM_antitox ParE_toxin E1_dh

JAIVJQ010000010.1:1-4518
Acidobacteria bacterium

Sigma70_ECF STY-kinase TPRs TPRs ? RelB-like ParE_toxin ?

JAIVIX010000033.1:1-3501
Verrucomicrobiaceae bacterium

? RelB-like ParE_toxin Amphi-Trp DDAH_eukar Cytochrom_C_asm

JAGQGS010000001.1:11426-18161
Thermomicrobiales bacterium

adh_short_C2 ? Uma2 Rieske Amidohydro_2 YjgF_endoribonc Aminotran_5 ParE_toxin PhdYeFM_antitox PhdYeFM_antitox

JACKEV010000008.1:705356-720207
Alphaproteobacteria bacterium

Peptidase_S9 Peptidase_S9_N Zn_ribbon_TFIIB Zn_ribbon_TFIIB Usp Creatininase Usp ? Cu-binding_MopE Epimerase Cupin_2 ? ParE_toxin ? Lambda_Bor FG-GAP FG-GAP FG-GAP FG-GAP FG-GAP_3 FG-GAP_3 MASE1 His-Kinase His-Kinase Acetylase ? ?

JAJHQX010000020.1:21514-28610
Armatimonadota bacterium

Methyltransf_4 Uma2 Amidase TPK_catalytic SteA-like_C MctB AAA_3 AAA_lid_2 ? ParE_toxin

JADLAP010000119.1:6760-14132
Phycisphaeraceae bacterium

? PhdYeFM_antitox ParE_toxin LacI Peripla_BP_3 N_methyl ? DUF4838 CBM9_1 ? GH123_N GH123_N LamininG LamininG

JADLAX010000027.1:67066-81765
Anaerolineae bacterium

PNPOx_N UPF0182 rve rve_3 HTH_Tnp_1 DinB_2 S-methyl_trans DUF1638 ParE_toxin PhdYeFM_antitox NUDIX Cytochrome_B Fer4_7 FlpD Cytochrom_C COX1 UbiA COX2

JAIORC010000072.1:1-2959
Phycisphaerae bacterium

PHP PALP HicB_lk_antitox HicA_toxin PhdYeFM_antitox ParE_toxin

QWGW01000019.1:23410-39738
Haliscomenobacteraceae bacterium CHB4

? TPRs Sigma70_r4 Sigma70_r2 OmpA DUF937 ? MRM3-like_sub_bind SpoU_methylase MazG MazG ParE_toxin Ribosomal_S15 ? RNase_PH RNase_PH_C PNPase RNase_PH RNase_PH_C KH_1 S1 PKD_4 Por_Secre_tail ? RoxA-like_Cyt-c CCP_MauG SnoaL

JACZUY010000110.1:1-7949
Proteobacteria bacterium

UDG PIN FA_desaturase SLT MoCF_biosynth G_glu_transpept ParE_toxin PhdYeFM_antitox DUF2064

JAKEFM010000023.1:1-11067
Planctomycetota bacterium

Fic PhdYeFM_antitox PIN Peptidase_M64 M64_N ParE_toxin PhdYeFM_antitox SLBB_2 SLBB Poly_export Glycos_transf_2 Glycos_transf_2 Glycos_transf_1 Glyco_transf_4 ? PMT_2

JAKEFD010000147.1:1-10457
Gemmataceae bacterium

STY-kinase HD_5 GAF_2 FHA DUF4139 ParE_toxin ? AP_endonuc_2 ? ? Peptidase_M48 Peptidase_M48_N Polysacc_deac_1

JAEMPV010000034.1:164-5757
Caldisericum sp.

Peptidase_M20 ABC2_membrane ABC_tran ? ParE_toxin RHH_1 Methyltransf_25 Peptidase_M3

JALHUF010000027.1:1-4525

Terriglobia bacterium

BipA_C EFG_C EFG_III GTP_EFTU_D2 ? HTH_3 HigB-like_toxin ParE_toxin ? ?

JALHWK010000648.1:1-9110
Sedimentisphaerales bacterium

ParB_N HTH_Tnp_1_2 LacAB_rpiB PhdYeFM_antitox ParE_toxin Terminase_6C HTH_3 Peptidase_M78 ? Hydrolase_4 GCE_fung ?

JALHYD010000467.1:742-4291
bacterium

HicA_toxin HicB_lk_antitox ? DUF7387 HicA_toxin ? RelB-like ParE_toxin Caspase

JALJPP010001998.1:1-1509

Rivularia sp. ALOHA_DT_140

ParA-Soj-NTPase ? ParE_toxin Antitox_RHH

JAGLZM010000205.1:355-6397
Candidatus Aegiribacteria sp.

Yip1 ? MgtE_N CBS CBS MgtE GlgB_N CBM_48 Alpha-amylase Alpha-amylase Alpha-amylase_C ? RelB-like ParE_toxin

JAGLZA010000201.1:4329-17517
candidate division WOR-3 bacterium

TEN_YD-shell ? ? F-box PIN ? RelB-like ParE_toxin RHS DUF192 DeoC SMC_N CHASE2 Guanylate_cyc B12-binding Radical_SAM NurA

JAGLTD010000388.1:1-4874
Spirochaetes bacterium

Cas_Cas7 Cas_TM1802 Polbeta ParE_toxin RelB-like Cas6b_N Cas6b_C CBM_6

JAGLHJ010001754.1:1-4873
Candidatus Pacearchaeota archaeon

AAA_23 Calcineurin ? DUF2238 MMR_HSR1 ? MazG ParE_toxin ?

JAKLJB010000021.1:43363-59252
Thermoanaerobaculia bacterium

RVT_3 Ala_racemase_N Rhomboid DivIVA Amidohydro_3 Amidohydro_3 FTCD_C ? ParE_toxin ? YfhO HHH_8 PHP DNA_pol_A 5_3_exonuc 5_3_exonuc_N ? ? DabA

JALNYY010000062.1:15628-22838
Methylobacter sp.

DUF364 DUF4213 ? Rhodanese ANKs ANKs Nitro_FeMo-Co DIMCO_N YunG ? ParE_toxin DUF6290 ? Nitroreductase

JAIQZM010000030.1:30945-45699
Argonema galeatum A003/A1

Peptidase_M23 Fer4_5 Fer4_5 AAA_lid_14 NtrC-AAA cNMPBD DUF488 DUF488 DuOB ? ? ParE_toxin PhdYeFM_antitox N6_N4_Mtase ? ? Uma2 PhdYeFM_antitox Met_synt_B12 B12-binding B12-binding_2 Pterin_bind S-methyl_trans

JALQCR010000001.1:123115-138132
Pseudanabaena sp. Salubria-1

Cytochrome_CBB3 RNase_PH_C RNase_PH TPRs Rubrerythrin PETase PETase DUF1400 FAD_binding_3 ? ParE_toxin ? Sigma70_r4 NACHT NCH2 ? TumE PsbM S1 S1 S1 ?

WRBW01000021.1:153-13866
Coriobacteriia bacterium

Acetylase PIG-L Cytochrom_C552 ? DUF6036 ? ParE_toxin DUF6290 PBP_N LytR_cpsA_psr DNA_methylase Endonuc-MspI PALP ACT_4 ILVD_EDD_N ILV_EDD_C KARI_N KARI_C

JAGTJV010000004.1:21802-32677
Gloeotrichia echinulata DEX184

DUF2281 PIN Uma2 CPSF100_C Uma2 DUF29 ? DUF2281 ParE_toxin DUF2281 PIN HTH_Tnp_4 DDE_Tnp_4 Glycos_transf_2 Acyl_transf_3 ? ?

JAGFJE010000251.1:1-17274
Phaeodactylibacter sp.

TetR_N FAD_binding_3 MarR NTPase_1 ? TonB_dep_Rec_b-barrel Plug CarbopepD_reg_2 ParE_toxin RelB-like Trans_reg_C Receiver His-Kinase His-Kinase OMP_b-brl_3 Plug CarboxypepD_reg ? OMP_b-brl_3 Plug CarboxypepD_reg PepSY_like

JAAELT010000279.1:24418-37223
bacterium

Receiver fn3 fn3 Reprolysin_4 Apiosidase Methyltransf_33 UB2H Transgly Transpeptidase HD_5 PhdYeFM_antitox ParE_toxin Glycos_transf_1 Glyco_transf_4 Amidohydro_3

JAAEQP010000826.1:3343-16903
bacterium

GFO_IDH_MocA DUF362 Methyltransf_25 PQQ_2 PQQ_2 PQQ_2 PQQ_2 PQQ_2 PhoD_N DUF2341 LamininG Hydrolase_6 Hydrolase_like HAD_2 PhdYeFM_antitox ParE_toxin UvrD_C UvrD_C UvrD-helicase BD-FAE

JANSXX010000012.1:6961-20611
bacterium

YdgH_BhsA-like FIMAH Peptidase_M74 Pyr_redox_2 cNMPBD His-Kinase Thioredoxin_4 Na_H_Exchanger TrkA_N ParE_toxin DUF6290 Abhydrolase_2 MFS_1 Glyoxalase Glyoxalase Peptidase_M48 Peptidase_M48_N ? PIN

JANWYZ010000003.1:581-17498
bacterium

QRPTase_N GlcNAc_2-epim Glyco_hydro_2_N2 Glyco_hydro_2 Glyco_hydro_2_C Mannosidase_ig Ig_mannosidase Peptidase_S41 PDZ_6 Tricorn_C1 PD40 PD40 DPPIV_N PD40 PD40 PD40 PD40 Redoxin ParE_toxin RelB-like PMT_2 Gly_kinase OMP_b-brl GcpE Ala_racemase_N Ala_racemase_C F_bP_aldolase

JAPLYB010000016.1:5675-16454
Candidatus Pacearchaeota archaeon

? DUF7670 tRNA_Me_trans NFACT-R_2 ? Rubrerythrin RUBY_RBDX Thioredoxin_3 DUF7557 ParE_toxin Fer4 SfLAP ? Prismane Prismane ? Phage_integrase Neur_chan_memb

JAPLNJ010000622.1:416-8445
Planctomycetota bacterium

ParE_toxin PhdYeFM_antitox ? ? ? Melibiase ? ?

JAPLOD010000808.1:1-7542
Planctomycetota bacterium

Beta-prop_ACSF4 PQQ_2 PQQ_2 TPRs zf-HC2 Sigma70_r2 Sigma70_r4_2 ? ParE_toxin PhdYeFM_antitox SRAP

JAPUJJ010000338.1:3904-4232
Acidobacteria bacterium

PhdYeFM_antitox ParE_toxin ? ? ?

JAQCNZ010000020.1:151015-159956
Chloroflexota bacterium

ABC_tran TOBE_2 BPD_transp_1 BPD_transp_1 SBP_bac_8 23S_rRNA_IVP Aminotran_3 ? ParE_toxin ? Lysyl_oxidase

JAQBYG010000020.1:7592-15136
bacterium

Ribosomal_S18 CtpB_N-like PDZ_6 Peptidase_S41 ? Trypsin PDZ_2 RF-1 PCRF PhdYeFM_antitox ParE_toxin PPDK_N HGTP_anticodon tRNA-synt_His

JAPTXZ010000265.1:400-2486
Actinomycetota bacterium

? PhdYeFM_antitox ParE_toxin F420H2_quin_red

JAPTXN010000039.1:201-6999
Actinomycetota bacterium

ParE_toxin PhdYeFM_antitox ? ? TetR_C_6 TetR_N Aconitase Aconitase Aconitase_C ?

JAOKSR010000015.1:1-13471
Candidatus Omnitrophota bacterium

NUDIX Ysc84 EipA ? ? ParE_toxin RHH_5 DUF134 Dus AAA_14 DUF4143 Peptidase_S8 Big TIG DUF4172 Fic AbiEii

JAJRDQ010000417.1:1-3137
Methylococcaceae bacterium

Cytochrom_D1 ParE_toxin DUF6290 Mrr_cat_2 Cytochrome_CBB3 Cytochrome_CBB3 Cytochrom_D1

JAQUAZ010000127.1:40-4937
Methylococcales bacterium

ParE_toxin ? ? ? Zot DUF5455 ? Phage_coatGP8

JAQUKL010000222.1:1-4223
bacterium

D5_N HTH_17 ? ParE_toxin PhdYeFM_antitox Phage_integrase

JAPWKE010000003.1:2905946-2923168
Acidobacteriaceae bacterium

Aminotran_3 AIRS_C AIRS Formyl_trans_N Peptidase_M50 Peptidase_M50 MerR_1 Vgb_lyase RtcB ParE_toxin ? EF_TS Ribosomal_S2 Ribosomal_S9 Ribosomal_L13 CarboxypepD_reg OMP_b-brl_4 OMP_b-brl_4 TonB_C

JAIEFT010000014.1:10-8544
Clostridiales bacterium

? ParE_toxin DUF6290 PCRF RF-1 DUF1385 PrmC_N MTS Ribosomal_L31 ADH_N ADH_zinc_N Peptidase_M23 ?

JARGFK010000237.1:151-1933
Desulfobacterales bacterium

HTH_23 Y1_Tnp ParE_toxin PhdYeFM_antitox

JAOTYS010000732.1:87-2064
Gammaproteobacteria bacterium

RHH_1 DUF6290 ParE_toxin Ysc84 ?

JASJEF010000275.1:1-8292

Xenococcaceae cyanobacterium MO_207.B15

RVT_N ACR_tran ? ? ParE_toxin TraY YtkA HlyD_D23 YtkA ? HTH_3 Gp49 Trans_reg_C Receiver

JASJDW010000001.1:17146-31227
Crocosphaera sp.

GST_C_2 GST_N_2 DUF6887 DUF6888 FitA-like_RHH ? PIN HicB_lk_antitox ParE_toxin PhdYeFM_antitox GDE_N GDE_C Peptidase_M16_C Peptidase_M16 Peptidase_M16 Receiver His-Kinase His-Kinase GAF PAS PAS GAF CBS CBS CBS CBS Coprogen_oxidas ?

JASJCT010000022.1:172960-192690
Myxococcota bacterium

Patatin PC4 AIRS_C AIRS AIRS_C AIRS FGAR-AT_linker GATase_5 HTH_17 ABC-ATPase CCDC113_CCDC96_CC AAA_15 HATPase_c_4 SLFN_AlbA_2 ParE_toxin PhdYeFM_antitox ? Ribonucleas_3_3 dsrm Thioredoxin_4 VKOR GreA_GreB GreA_GreB_N RecG_dom3_C Helicase_C DEAD RecG_wedge HEATs SGSH_C Sulfatase

SZPN01000101.1:335-7290
Acidobacteria bacterium ACD

ParE_toxin ? Phage_integrase Zn_ribbon_DnaG Toprim_2 HTH_3 ?

JAUSCR010000025.1:5790-5711
Dehalococcoidia bacterium

BPD_transp_1 E1_dh Transket_pyr Transketolase_C RelB-like ParE_toxin Biotin_lipoyl E3_binding 2-oxoacid_dh NAD_kinase

JAURYR010000964.1:283-6781
Methylobacter sp.

? ParE_toxin DUF6290 DNA_pol3_chi Peptidase_M17_N Peptidase_M17 LptF_LptG LptF_LptG

JASLOA010000041.1:31020-47331
Kiritimatiellia bacterium

Tyr_Deacylase ADK SurA_N_3 Rotamase_3 Kdo ABC_tran ABC_membrane Glyco_hydro_2_N2 Glyco_hydro_2 Glyco_hydro_2_C RelB-like ParE_toxin A-2_8-polyST PS_pyruv_trans Glycos_transf_2 ? Glyco_transf_9 Glyco_transf_9 PfkB

JALYJY010000030.1:2046-8534
Pseudomonadota bacterium

HAD CutA1 Mur_ligase_C Mur_ligase_M MurD-like_N DUF4197 ParE_toxin PhdYeFM_antitox SufBD_N SUFBD_core ? ? DAHP_synth_1 tRNA-synt_2 Elong-fact-P_C EFP EFP_N

JALYUK010000495.1:1-2739
Actinomycetota bacterium

APH ? PhdYeFM_antitox Unstab_antitox ParE_toxin ?

JALZIH010000398.1:1-4809
Acidobacteriota bacterium

? ? ? ParE_toxin ? Peptidase_M15 EPSP_synthase SKI

JALZIH010000441.1:1-3749
Acidobacteriota bacterium

YidC_periplas YidD Ribonuclease_P Ribosomal_L34 ParE_toxin ? Formyl_trans_N CarboxypepD_reg ?

JAISJO010000023.1:1-8626
Propionibacteriaceae bacterium

BRCT ParE_toxin PhdYeFM_antitox Phage_integrase SBP_bac_8 Alpha-amylase ABC_tran DrrA1-3_C ? Zn_ribbon_9 CT398_CC

JAIRYI010000100.1:1-11127
Desulfarculales bacterium

? 4HBT_2 Nitroreductase CBS CBS CorC_HlyC LNT_N CN_hydrolase DUF6290 ParE_toxin ADH_zinc_N ThiF ADH_N IspD Ribul_P_3_epim CTP_transf_3 NeuB Hexapep Radical_SAM

JAITCS010000155.1:1135-6045
Cellulomonadaceae bacterium

FAD_syn ? ParE_toxin PhdYeFM_antitox GerE ? ?

JAITOZ010000260.1:1-7922
Planctomycetaceae bacterium

T2SSE FMN_red Beta-prop_THOC3 SHD1 DUF7382 ? ParE_toxin ? DUF4139

JAYEMF010000019.1:15315-23086
Campylobacterota bacterium

FIST FIST_C PAS His-Kinase His-Kinase PAS HAMP_2 MCPsignal PAS GAF PAS His-Kinase His-Kinase Receiver GerE RHH_5 ParE_toxin HepT-like

JAYKJI010000224.1:725-7592
Snowella sp.

WYL WCX Cas6 Cas3_HD DEAD Cas3-like_C_2 ? DevR RHH_4 ParE_toxin Cas_Cas5d

JAYKJI010000431.1:5001-15257
Snowella sp.

STY-kinase FGS 23S_rRNA_IVP ? ? BrnT_toxin BrnA_antitoxin AIPR ParE_toxin DUF2281 XisI STY-kinase FGS Fer4_12 ?

JAYKGY010000050.1:932-7709
Snowella sp.

Cas3_HD DEAD Cas3-like_C_2 ? BrnT_toxin Cas8b_N Cas8b_C DevR Cas_Cas5d RHH_1 ParE_toxin

JAWQFP010000179.1:4547-4383
Scytonema sp. PMC 1069.18

PIN Uma2 ? ParE_toxin ? Calcineurin ? ?

JAWQFX010000217.1:1-8025
Oscillatoria sp. PMC 1076.18

Met_synt_B12 B12-binding B12-binding_2 Pterin_bind S-methyl_trans PhdYeFM_antitox ParE_toxin Pps Pps Pps Pps Pps Pps Pps Pps Pps Pps Pps Pilin_GH LTXXQ ?

QGUM02000135.1:1-4491
Actinomycetes bacterium

Sod_Fe_C Sod_Fe_N ParE_toxin PhdYeFM_antitox adh_short HSDR_N_2 Cyt_c_ox_IV COX1

VOGC01000005.1:1006-2836
Candidatus Weimeria bifida

ParE_toxin RHH_3 Rep_OBD

SIAS01000005.1:44330-50050
Verrucomicrobiia bacterium

? Thioredoxin Cys_rich_CPXG DUF2064 GHMP_kinases_C GHMP_kinases_N Ldh_1_N Ldh_1_C ? ParE_toxin ANKs ANK_LRRK2 ANKs

WLIA01000003.1:13853-25332
Actinomycetota bacterium

UbiA SprT-like Acetylase ? Cys_Met_Meta_PP Aminotran_1_2 GATase PhdYeFM_antitox ParE_toxin TauE APH ? Abhydrolase_3 AhpC-TSA ? Whib

WJFC01000001.1:257368-273595
Scytonema sp. UIC 10036

? TPS PIN ? ? TPRs TPRs TPRs RNaseH-CATRA+Caspase ? ParE_toxin DUF928 XisI Antibiotic_NAT Fer2 Fer2_2 FAD_binding_5 CO_deh_flav_C Ald_Xan_dh_C MoCoBD_1 MoCoBD_2 Coq4

WJFC01000015.1:85327-97219
Scytonema sp. UIC 10036

MCPsignal HAMP TPR_CcmH_CycH CheR CheR_N CheW ? ? ? ? ParE_toxin ? DUF6972 DUF1016_N DUF1016_N YhcG_C Gp49 ? His-Kinase Uma2

VXXA01000172.1:1044-8959
Gemmatimonadetes bacterium

AHSA1 HTH_20 CarbopepD_reg_2 Plug TonB_dep_Rec_b-barrel ? DUF3667 ? DUF6290 ParE_toxin

JAAAOF010000278.1:48-7279
Bacteroidetes bacterium

Receiver HTH_18 Receiver His-Kinase His-Kinase Y_Y_Y Reg_prop Reg_prop Reg_prop Reg_prop Reg_prop Reg_prop ParE_toxin PhdYeFM_antitox Phage_integrase

SAAS01000097.1:733-1603
Clostridia bacterium

ParE_toxin ? ?

JAAHGY010000434.1:42-5924
Cyanothece sp. SIO2G6

? PhdYeFM_antitox ParE_toxin PPC TPRs TPR_NPHP3 TPR_MalT TPRs TPRs TPRs TPRs TPRs RNaseH-CATRA+Caspase

JAAKFX010000055.1:1-10326
Candidatus Anoxychlamydiales bacterium

HTH_16 RNA_pol_Rpb2_6 RNA_pol_Rpb2_7 RNA_pol_Rpb1_1 RNA_pol_Rpb1_2 RNA_pol_Rpb1_2 RNA_pol_Rpb1_3 RNA_pol_Rpb1_4 RNA_pol_Rpb1_5 MFS_1 ? TAL_FSA PhdYeFM_antitox ParE_toxin ?

JAACEL010000445.1:197-3689
Nitrospiraceae bacterium

peroxidase peroxidase ParE_toxin RelB-like

JAAUTN010000243.1:1-5656

Leptolyngbyaceae cyanobacterium SM1_4_3

TPRs TPRs ? ParE_toxin UPF0150 DUF2283 DAO DAO FA_desaturase ? EAL GGDEF

JAAUUQ010000134.1:5058-6680
bacterium

NACHT HEATs RelB_like_antitoxin ? ? ParE-like_toxin DUF2281 ParE_toxin CTP_transf_1 RRF DDE_Tnp_IS1

JAAUPY010000362.1:2799-2041
Anaerolineae bacterium

DEAD PriA_C Helicase_C Zn_ribbon_PriA ParE_toxin RelB-like

JAAUSF010000103.1:2859-2862

Cyanobacteria bacterium RU_5_0

HicA_toxin ? ParE_toxin ? DUF2283 ? ? DUF2283 HicB_lk_antitox UPF0150

JAAUOH010000086.1:955-7517

Pleurocapsa sp. CRU_1_2

? ParE_toxin RelB-like DUF4926 NTP_transf_2 SQHop_cyclase_C SQHop_cyclase_N MMR_HSR1 YchF-GTPase_C Receiver Trans_reg_C OKR_DC_1_C OKR_DC_1

JAAYNH010000010.1:16492-26719
Treponema sp.

ACT_AHAS_ss ALS_ss_C Sirtuin WKF Fer4_5 Fer4_5 Fer4_18 Lipase_3 ABC_tran ABC_tran ? ParE_toxin ? Gp49 MazE_antitoxin HTH_3 Sel1 Sel1 DUF1538 DUF1538 P-II P-II

JAAYDF010000030.1:6594-9825
Phycisphaerae bacterium

? Zn_ribbon_recom Resolvase Phage_integrase Phage_integrase ParE_toxin PhdYeFM_antitox ? ? ArsP_1 ? CCG CCG LutB_C Fer4_8 LUD_dom

JAAZNU010000021.1:110343-122167
Veillonellaceae bacterium

Flg_bb_rod FlgK_D1 Flg_bbr_C Flagellin_N DUF6470 FliW CsrA Flagellin_N Flagellin_C DUF6290 ParE_toxin FlaG FliD_N Flagellin_IN FliD_C FliS ? ? zf_Tbcl zf_Tbcl_2 ?

WTAW01000196.1:1-5384
Candidatus Aminicenantes bacterium

Pps Pps Acetylase TOP_N Peptidase_M3 Antitox_RHH ParE_toxin SEC-C

JABFSO010000050.1:1-7949
Cyclobacteriaceae bacterium

Peptidase_M3 Peptidase_M3 ParE_toxin RelB-like MarR YceI CBM9_1 DUF5916 MTS Fe-ADH ADH_Fe_C AbiEii AbiEii

JABFSN010000010.1:5066-8050
Phycisphaerales bacterium

AcetylCoA_hyd_C Acetylase DUF4062 Sel1 Sel1 Sel1 Sel1 Sel1 Sel1 Sel1 Sel1 Sel1 Sel1 Sel1 Sel1 PhdYeFM_antitox ParE_toxin DUF7560 ?

JAADIN010000091.1:1863-5155
Planctomycetota bacterium

PhdYeFM_antitox ParE_toxin Phage_integrase ? UPF0102 Ribosomal_L19 tRNA_m1G_MT Ribosomal_S16 ParB_C

JABMQP010000080.1:113-8403
Candidatus Aminicenantes bacterium

? Methyltransf_31 HTH_3 RHH_3 ParE_toxin HicB_lk_antitox HicA_toxin NTP_transf_2 PRiA4_ORF3 RNaseH-CATRA+Caspase Beta-lactamase DUF554

JABMMJ010000277.1:1-2860
bacterium

Ribosomal_L27 OpcA_G6PD_assem OpcA_G6PD_C ParE_toxin RelB-like G6PD_C

JABUFB010000009.1:223317-243948
Armatimonadetes bacterium

Put_Phosphatase PMT Hepar_II_III_C Hepar_II_III_N CBM_11 GFO_IDH_MocA_C3 GFO_IDH_MocA ? Acetylase Rab11-FIP3 Zn_Ribbon_1 ParE_toxin DUF5696 Amidohydro_2 3keto-disac_hyd BNR_2 Amidohydro_2 PAP2 CBM9_1 Glyco_hydro_20b Glyco_hydro_20 DUF3429

JAAXWU010000099.1:1-7524
Chlorobaculum sp.

? GATase SKI DHQ_synthase DHQS_C RHH_1 ParE_toxin ? Chorismate_synt ALAD

MEKY01000062.1:15241-12654

Acidobacteria bacterium RIFCSPLOWO2_12_FULL_59_11

? ? 5_3_exonuc_N 5_3_exonuc DNA_polI_exo1 DNA_pol_A PIN PhdYeFM_antitox ParE_toxin LolA_like ? ABC_tran FtsX MacB_PCD LolA_like ? CarboxypepD_reg Plug OMP_b-brl_4

MEXX01000024.1:2279-13101

Candidatus Aminicenantes bacterium RBG_13_62_12

Polysacc_deac_1 PemK_toxin DUF676 OMP_b-brl_2 DcuC Peptidase_M20 ParE_toxin PhdYeFM_antitox TPRs PIN PhdYeFM_antitox AhpC-TSA AhpC-TSA ? Acyltransferase

MGBU01000112.1:9275-9273

Candidatus Rokubacteria bacterium GWA2_73_35

ParE_toxin ? DUF6941 EamA EamA AFOR_N AFOR_C TPRs TPRs TPRs TPRs TPRs TPRs TPRs TPRs TPRs TPRs TPRs TPRs Hydrolase

MGPV01000013.1:25807-30850

Deltaproteobacteria bacterium GWC2_65_14

HicB_lk_antitox PIN PIN MazE_antitoxin ? PIN RHH_1 ParE_toxin PhdYeFM_antitox AAA_14 DUF4143

MGRF01000023.1:459-6821

Deltaproteobacteria bacterium RBG_19FT_COMBO_60_16

Na_H_Exchanger DUF3303 ParE_toxin PhdYeFM_antitox ORF6N DoxX HTH_3 ? His_biosynth

MGST01000313.1:16361-28618

Deltaproteobacteria bacterium RIFOXYA12_FULL_58_15

AbiEii SNase LTD SNase Fer4_20 Pyr_redox_2 DHO_dh Fer4_21 His-Kinase PilZ ? ParE_toxin XPB_DRD ResIII Helicase_C DUF790 PPDK_N PhdYeFM_antitox

MHGP01000024.1:10174-19657

Omnitrophica WOR_2 bacterium RIFCSPHIGHO2_02_FULL_48_11

23S_rRNA_IVP ? 23S_rRNA_IVP HTH_3 Cupin_2 23S_rRNA_IVP Receiver PhdYeFM_antitox HisKA_7TM His-Kinase His-Kinase His-Kinase His-Kinase 23S_rRNA_IVP ParE_toxin

MHGK01000072.1:5823-5820

Omnitrophica WOR_2 bacterium RIFCSPHIGHO2_01_FULL_48_9

RHH_3 PIN ParE_toxin SNase LexA_DNA_bind Peptidase_S24 Radical_SAM DUF748 DUF3568 ?

MHXR01000049.1:53153-67983

Planctomycetes bacterium GWA2_50_13

CN_hydrolase NAD_synthase ? HD ? Cas_Cas1 CRISPR_Cas2 ? ParE_toxin Acyltransferase S1 S1 S1 S1 S1 S1 Sigma70_r4 Sigma70_r2 Sigma70_r1_2 Pribosyltran Peptidase_S24 HTH_3 Amidohydro_3 Cytidylate_kin

MKUL01000057.1:210-12161
Alphaproteobacteria bacterium 41-28

TraF ParE_toxin HicB TraH TraG_N TrwB_AAD_bind TrwB_AAD_bind Sel1 Sel1 Sel1 Sel1 ? TPRs

LWDK01000048.1:38915-51134

Proteobacteria bacterium SG_bin7

ICMT ? ? ? ? RWP-RK Recombinase PAD ParE_toxin RH3_dom PSCyt1 YceI ? ? ? CRCB

NBMI01000049.1:27722-44302

Candidatus Parabeggiatoa sp. nov. 2

RuBisCO_small CbiA LysM SH3_3 ? ? ? ParE_toxin ? ? AAA_15 DUF4435 HSP70 Nuf2_DHR10-like Fer4_12 FGS

PCVV01000055.1:1-1245

Candidatus Pacearchaeota archaeon CG11_big_fil_rev_8_21_14_0_20_30_13

? DUF7557 ParE_toxin Dockerin_1

PHEO01000147.1:1-7591
Alphaproteobacteria bacterium HGW-Alphaproteobacteria-11

GerE Abhydrolase_6 PATR Autotransporter HTH_31 ? RHH_1 ParE_toxin Lactamase_B HTH_5

NMQD01001547.1:1-4869
Fischerella thermalis CCMEE 5319

T2SSE T2SSE T2SSF T2SSF ? ParE_toxin ? ?

PVWL01000028.1:45416-55668
Cyanosarcina cf. burmensis CCALA 770

Xre_MbcA_ParS_C Xre-like-HTH RES ? HTH_23 ? ? RHH_1 ParE_toxin PhdYeFM_antitox HTH_26 ? Polbeta HEPN ? ?

PXSZ01000044.1:73375-82073

Bacteroidetes bacterium QH_10_64_37

Proton_antipo_M Oxidored_q5_N Pterin_4a Proton_antipo_M ? Aldedh Isochorismatase HicB ParE_toxin RelB-like DUF5615 DUF433 DUF433

PXTI01000061.1:1609-11573

Bacteroidetes bacterium QH_8_67_23

SecD_SecF_C Sec_GG STAS Metallophos_2 Amidohydro_1 ParE_toxin PhdYeFM_antitox Peptidase_M48 Rce1-like LysM LysM LysM DPBB_1 TPRs Trehalase Mannitol_dh Mannitol_dh_C

QHSU01000071.1:53446-68487
Candidatus Rokuibacteriota bacterium

? SLBB SLBB SLBB SLBB SLBB_2 Poly_export PhdYeFM_antitox AMP-binding DegT_DnrJ_EryC1 ParE_toxin DegT_DnrJ_EryC1 NAD_binding_5 Lycopene_cycl DAO FAD_binding_2 GNVR Wzz Bac_transf GDP_Man_Dehyd Glyco_trans_2_3 Glycos_transf_2 PhdYeFM_antitox RelB-like

QEVQ01000161.1:1-4807
Proteobacteria bacterium

Memo ? PhdYeFM_antitox ParE_toxin ?

QMXR01000120.1:255-3220
Candidatus Bathyarchaeota archaeon

? PIN MazE_antitoxin PIN VAPB_antitox RHH_3 ParE_toxin UPF0175

RFIY01000040.1:13182-26060
Candidatus Dadabacteria bacterium

? SpoVG Receiver PilZ Na_Ala_symp BATS Radical_SAM IstB_IS21 RHH_1 ParE_toxin HAD Radical_SAM HrcA_DNA-bdg HrcA GrpE Peptidase_M20 Acetylase Pribosyltran Pribosyltran_N

RDXI01000010.1:38673-51877
Bacteroidetes bacterium

GldL_N FGS PorP_SprF HEM4 DUF4271 ? SUFBD_core SufBD_N ParE_toxin ? HTH_3 Methylase_S Methylase_S Big Big Por_Secre_tail

SHNH01000008.1:41157-52305
Candidatus Saccharibacteria bacterium

? Glyco_hydro_18 fn3 5_3_exonuc_N 5_3_exonuc DNA_pol_A Hydrolase MazE_antitoxin ? Antitox_RHH ParE_toxin ? Aminotran_1_2 Methyltransf_25 NUDIX PhdYeFM_antitox PIN ?

SMKE01000128.1:1-5658
Micromonospora fluostatini

? ? NUDIX ParE_toxin PhdYeFM_antitox GntR ? ? ? HD

SMYC01000093.1:4226-19990
Acidobacteria bacterium

DUF4159 DUF6483 ? Trans_reg_C Sulfatase Sulfatase Beta-lactamase Metal_resist DUF6290 ParE_toxin Cu-oxidase_2 Cu-oxidase Cu-oxidase_3 ? ? Amidohydro_1 Abhydrolase_1 STY-kinase PEGA PEGA TonB_C Peripla_BP_2 ?

RDSB01000021.1:64779-72306
Marinifilum sp. JC120

Lar_restr_allev Phage_CP76 ? ? HTH_3 HTH_26 Peptidase_S24 ? ParE_toxin ? ? ? ? Lar_restr_allev PAPS_reduct ?

VBOP01000117.1:3425-3422
Nitrospirae bacterium

MazE_antitoxin PemK_toxin ParE_toxin PhdYeFM_antitox FecR

VIKD01000012.1:1-8939
Planctomycetota bacterium

Calx-beta Calx-beta ? BD-FAE ? ? HTH_3 ParE_toxin PhdYeFM_antitox ? MmeI_Mtase Peptidase_M78 ? ?

NZ_CM001794.1:1677599-1694393
Treponema denticola H1-T

? Aldedh PDDEXK_2 ParE_toxin HAMP Guanylate_cyc FecR ? ? LRRs LRRs BrnT_toxin BrnA_antitoxin HAD Colicin_V DNA_pol3_delta2 Glyco_transf_28 Glyco_tran_28_C

NZ_ANKQ01000001.1:1510034-1527054
Microcystis aeruginosa TAIHU98

Methylase_S Methylase_S Ribosomal_L27A NTP_transf_2 ? SIS SIS GATase_6 Ribosomal_L6 Ribosomal_L6 Ribosomal_L18p ParE_toxin N6_Mtase HsdM_N Ribosomal_S5_C Ribosomal_S5 ? REase SecY ADK Tnp_DNA_bind DDE_Tnp_1 Dimer_Tnp_Tn5

NZ_JABCUQ010000045.1:1-7102
Mobiluncus mulieris

PhdYeFM_antitox ParE_toxin Fapy_DNA_glyco H2TH zf-FPG_IleRS ? ? ? Amidohydro_1 ?

NZ_CAUSPZ010000007.1:3887-27650
Rothia mucilaginosa

Aldo_ket_red PhdYeFM_antitox p450 Cu-oxidase_3 Cupredoxin_1 GT87 Rce1-like New_glue GT87 SLT ParE_toxin Sugar_tr ABC_tran ABC_tran MTHFR Meth_synt_2 Meth_synt_1

NZ_JAVIIS010000011.1:25453-40409
Mesorhizobium australafricanum

PP-binding adh_short_C2 Acyl_transf_3 Glycos_transf_2 Glyco_tranf_2_2 Glycos_transf_2 ABC_membrane ABC_tran Sulfatase Acyl_transf_1 RHH_1 ParE_toxin ketoacyl-synt Ketoacyl-synt_C YceG YicC-like_N YicC-like_C Guanylate_kin ParE_toxin

NC_021184.1:3914077-3932840
Desulfoscipio gibsoniae DSM 7213

? Toprim Topoisom_bac Toprim_Crpt Toprim_Crpt ? AbiEii AbiEi_4 ? ParE_toxin ? Transposase_31 AMP-binding_C_2 Acetone_carb_G Hydantoinase_B Hydant_A_N Hydantoinase_A Hydant_A_C GAF_2 GGDEF_2 HTH_30 ?

NZ_BAHD01000004.1:149258-170430
Kineosphaera limosa NBRC 100340

adh_short_C2 DUF1949 UPF0029 Cys_Met_Meta_PP Viral_helicase1 UvrD-helicase PNKP_ligase Calcineurin AAA_33 PhdYeFM_antitox ParE_toxin Rce1-like DUF480 Methyltransf_25 Catalase Catalase-rel FUR DUF3418 OB_NTP_bind HA2_C HA2_N Helicase_C DEAD DUF1206 DUF1206 DUF1206 Methyltransf_12 Hen1_L PAC2

NZ_JAPTYE010000069.1:304-5195
Cylindrospermopsis raciborskii PAMP2011

? ParE_toxin Uma2 ? ? ? ? Eco57I

NZ_KV794747.1:62044-74536
Corynebacterium sp. HMSC070E08

ABC_tran Aminotran_5 ABC2_membrane DUF4288 STAS Sulfate_transp ADH_N ADH_zinc_N Pyr_redox_2 ParE_toxin Sirtuin DUF1648 MoaE MoCF_biosynth PhdYeFM_antitox ThiF Rhodanese PhdYeFM_antitox

NC_013851.1:1947865-1962385
Allochromatium vinosum DSM 180

FecCD ABC_tran Plug TonB_dep_Rec_b-barrel ? Arm-DNA-bind_3 Phage_int_M Phage_integrase RelB-like ParE_toxin ? HTH_17 HTH_17 ? DUF3987 ? SSB Cytochrom_C

NZ_JANIBK010000028.1:36874-6536
Methylomonas rivi

? polyprenyl_synt Exonuc_VII_S ParE_toxin DUF6290 ParE_toxin ? ? ? ? NERD Zn_ribbon_Top1 ? DXP_synthase_N

NZ_JADQUA010000014.1:8758-26941
Corynebacterium diphtheriae bv. mitis

GMC_oxred_C GMC_oxred_N LGFP Amidase_2 RHH_1 ABC_tran AAA_15 ABC-ATPase ABC-ATPase ABC_tran Methylase_S BCCT NAD_binding_8 GLF ParE_toxin Peripla_BP_2 FecCD AbiEii RepL AbiEi_2 RloB

NC_018012.1:1679493-1693944
Thiocystis violascens DSM 198

HTH_3 ? YgjP-like REase SWI2_SNF2 T1R_D3-like ParE_toxin RelB-like Methylase_S HsdM_N N6_Mtase RHH_5 ? ? Arm-DNA-bind_3 Phage_integrase Phage_int_M DUF6941 ?

NC_019680.1:1141688-1159388
Synechococcus sp. PCC 6312

? ? ? ? CRISPR_Cas2 ParE_toxin ? Cas_Cas5d DevR ? Cas3_HD DEAD Cas3-like_C_2 PIN ? DUF4957 Cas_Cas4 Cas_Cas1

NC_019682.1:5280596-5293917
Calothrix sp. PCC 7507

? RelB_like_antitoxin LRAT RecG_wedge DEAD Helicase_C RecG_dom3_C ? EF-Ts_N EF_TS Ribosomal_S2 ParE_toxin ? Glycos_transf_2 Glyco_trans_2_3 NADAR DUF6464 TPRs TPRs TPRs TPRs Pribosyltran ?

NC_019779.1:3408635-3423070
Halothece sp. PCC 7418

DNA_methylase ? S-methyl_trans Pterin_bind B12-binding_2 B12-binding Met_synt_B12 PAF-AH_p_II Abhydrolase_1 DUF1400 DHQS_C Fe-ADH Aminotran_1_2 Methyltransf_25 ParE_toxin ? HicB_lk_antitox HicA_toxin ? T5orf172 CopG_antitoxin ?

NC_012804.1:1040168-1051183
Thermococcus gammatolerans EJ3

ABC-3 RNase_HII GT87 Methyltransf_15 ? ParE_toxin eRF1_1 eRF1_2 eRF1_3 HEPN NTP_transf_2 AIRS_C AIRS QueC GPDPase_memb FeoA Creatinase_N Peptidase_M24 PUMA_CC

NZ_KB235948.1:3848664-3864153
Kamptonema formosum PCC 10802

GATase_7 Asn_synthase DAGK_cat YegS_C PAS PAS PAS PAS T4_CASP His-Kinase PhdYeFM_antitox ParE_toxin PhdYeFM_antitox DUF6290 Met_synt_B12 B12-binding B12-binding_2 Pterin_bind S-methyl_trans ? YafQ_toxin Ead_Ea22 ACP_PD ? DUF433 MotA_ExbB TOP6A-Spo11_Toprim

NZ_JAJNBZ010000027.1:5246-23883
Paenibacillus profundus

FecCD FecCD HlyD_2 Peripla_BP_2 ? ? ParE_toxin Asp_protease_2 ? HTH_18 Peripla_BP_2 N-OB tRNA-synt_2 Abhydrolase_1 Abhydrolase_1 SBP_bac_1 FlgM Beta-lactamase

NZ_ATXH01000022.1:27533-40052
Thermodesulfatator atlanticus DSM 21156

Lactamase_B_2 ? Hydrolase Sirtuin ? Fic ? ParE_toxin DUF6290 tRNA_synt_2f DALR_1 tRNA-synt_2e ? Peptidase_M24 Creatinase_N ABC_tran AAA_16 UvrA_inter DFP Flavoprotein

NZ_ATWH01000005.1:33040-46544
Glaciibacter superstes DSM 21135

AP_endonuc_2 GFO_IDH_MocA GFO_IDH_MocA_C DUF6807 SBP_bac_8 BPD_transp_1 PhdYeFM_antitox ParE_toxin BPD_transp_1 SBP_bac_3 Histidinol_dh HTH_1 LysR_substrate OCD_Mu_crystall PEP_mutase ABC_tran BPD_transp_1

NZ_JAFN01000001.1:2969799-2983804
Deferrisoma camini S3R1

DDE_Tnp_1 Nterm_IS4 DGCcoil GGDEF GalP_UDP_transf GalP_UDP_tr_C PGK TIM SecG Arm-DNA-bind_3 Phage_int_M Phage_integrase DUF6290 ParE_toxin ? ParE_toxin ? HTH_17 Zn_Ribbon_Prim DUF7146 Toprim_3 DUF3987

NZ_JIAG01000001.1:328022-342820
Haematomicrobium sanguinis DSM 21259

MFS_1 MFS_1 FemAB Acetyltransf_6 DoxX PhdYeFM_antitox DUF6290 ParE_toxin SAICAR_synt GARS_C GARS_A GARS_N ? UxaC Peripla_BP_3 LacI AP_endonuc_2 IclR_C HTH_IclR Sugar_tr GFO_IDH_MocA GFO_IDH_MocA_C3

NZ_AVFT01000052.1:20641-34693
Planktothrix agardhii NIVA-CYA 34

ParE_toxin DUF6384 Aminotran_5 RNase_H_2 Phytase-like HSP33 ? DUF4276 ABC-ATPase Helicase_C AAA_16 Acetyltransf_7 Acetylase Rv2175c_wHTH

NZ_ATYW01000002.1:367686-384492
Glycomyces tenuis DSM 44171

FMN_red ? BPD_transp_2 CarboxypepD_reg BPD_transp_2 ParE_toxin PhdYeFM_antitox adh_short ADH_N ADH_zinc_N MLTR_LBD HTH_31 Peripla_BP_6 ABC_tran Cas12f1-like_TNB OrfB_IS605 MFS_1 Trans_reg_C BTAD AP-ATPase TPR_MalT TPR_MalT ABC_tran BCA_ABC_TP_C

NZ_JIAI01000037.1:6196-12348
Pseudonocardia acaciae DSM 45401

? FitA-like_RHH PhdYeFM_antitox PIN ParE_toxin YoeB_toxin PhdYeFM_antitox PIN MazE_antitoxin Pps Pps MazE_antitoxin ? PhdYeFM_antitox ?

NZ_AUHN01000005.1:330197-347523
Jonesia quinghaiensis DSM 15701

PhdYeFM_antitox ParE_toxin adh_short_C2 adh_short IspD PS_pyruv_trans ? Synaptojanin fn3 Glyphos_transf Glyphos_transf Glyphos_transf ? Hexapep ? ABC2_membrane Gly_transf_sug ABC_tran

NZ_PCOW01000001.1:195675-215390
Microbacterium sp. MYb64

DUF1918 UPF0176_N Rhodanese Rhodanese_C VTT_dom DUF883_C ResIII AAA_lid_2 AAA_3 PhdYeFM_antitox ParE_toxin DUF222 HNH_2 ? Hydrolase Glyco_hydro_65N Glyco_hydro_65m Glyco_hydro_65C DNA_pol3_delta2 DNAX_ATPase_lid DNA_pol3_gamma3 RecR_HhH RecR_ZnF Toprim_4 RecR_C Transglutaminase TgpA_N vWA

NZ_LDYE01000004.1:44217-13137
Corynebacterium renale

? LeuA_dimer IPMS_D2 HMGL-like ParE_toxin PhdYeFM_antitox MurT_C Mur_ligase_M ? ? ? SpaA RNase_T

NZ_JAJVCN010000002.1:372797-387454
Kibdelosporangium philippinense

Cpn10 SBP_bac_3 BPD_transp_1 ? PIN Acetylase ParE_toxin PhdYeFM_antitox NHL NHL NHL Thioredoxin_8 VanW PG_binding_4 PG_binding_4 Acyl-CoA_dh_M Acyl-CoA_dh_1 Acyl-CoA_dh_C Receiver GerE DUF6191 Polyketide_cyc2 ABC_tran

NZ_BAAIFB010000002.1:20853-38844
Desulfosarcina cetonica

DUF6290 LMWPc CesT_Tir_1 ? RNaseH-CATRA+Caspase CHASE2 ParE_toxin HlyD_D23 Biotin_lipoyl_2 FecR PASTA PASTA PASTA PASTA PASTA PASTA PASTA MFS_1 Aldedh Nitroreductase MS_channel_1st MS_channel_2nd MS_channel_3rd PspA ? ParD_antitoxin

NZ_CP189962.1:2663888-2674294
Methylomonas sp. MS20

Rubredoxin NiFe-hyd_HybE ? HypA HupH_C HupH_C CTX_RstB Phage_coatGP8 ? DUF5455 Zot ? ? ParE_toxin ? NiFeSe_Hases

NZ_PUGE01000001.1:564028-586316
Corynebacterium sp. J010B-136

DUF2516 ? AMP-binding_C AMP-binding AMP-binding_C AMP-binding AAA_16 FCD GntR AA_permease Ketoacyl-synt_C ketoacyl-synt Fas_alpha_ACP Acyl_transf_1 MaoC_dehydratas DNA_pol_B_N Fas1-AflB-like_hel FAS_AT_central ? ParE_toxin PhdYeFM_antitox DUF2505 LmeA FAD_binding_4 MurB_C ?

NZ_LUUI01000127.1:101676-108783
Methylomonas lenta

DUF3368 HTH_3 Peptidase_M78 Methylase_S ? ParE_toxin Cation_efflux MerR-DNA-bind MerR_1 RelB-like Methylase_S ? REase SWI2_SNF2

NZ_JYIJ01000008.1:687-17911
Carbonactinospora thermoautotrophica

PhdYeFM_antitox ParE_toxin Phage_int_SAM_1 Phage_integrase LPP WXG100_2 ? ? Methyltransf_25 Eco57I ResIII Helicase_C ParA-Soj-NTPase

NZ_CP014209.1:2483637-2495625
Isoptericola dokdonensis DS-3

MFS_1 Multi_Drug_Res Multi_Drug_Res GyrI-like VTT_dom PhdYeFM_antitox ParE_toxin PspC PAP2 SSF TetR_N Catalase-rel Catalase RDD DUF4235 PAP2

NZ_JAJENT010000887.1:804-5194
Phormidium sp. CCY1219

DUF2281 UPF0175 PIN DUF3368 ? ? ParE_toxin ? ? PIN DUF3368 ? UPF0175 vATP-synt_AC39

NZ_JACKZR010000813.1:23827-40657
Nostoc sp. UCD120

DUF3531 NatA_aux_su HemY_N NUDIX UPF0150 ? ParE_toxin ? UvrB_inter CarD_TRCF_RID DEAD Helicase_C TRCF CorC_HlyC CBS CBS CNNM Aldo_ket_red AMP-binding AMP-binding_C PP-binding Condensation AMP-binding AMP-binding_C PP-binding Thioesterase HicA_toxin ?

NZ_FQUS01000045.1:11796-7532
Fodinibius roseus

PhdYeFM_antitox ParE_toxin UDPG_MGDP_dh_C UDPG_MGDP_dh UDPG_MGDP_dh_N DUF2442 DUF4160 Epimerase_2 ? Glycos_transf_4 Glyco_tran_WecG

NZ_LT828587.1:183-2037
Desulfamplus magnetovallimortis

ParE_toxin TraY ? HTH_37 ?

NZ_MTAV01000156.1:18933-27241
Nostoc sp. T09

Gly-zipper_Omp UPRTase Amino_oxidase Aldo_ket_red AMIN Beta-lactamase2 ? ParE_toxin XisI DJ-1_PfpI XisH

NZ_NOLI01000095.1:72-13699
Nostoc sp. 'Peltigera membranacea cyanobiont' 213

ResIII Helicase_C AAA_23 Laminin_II Uma2 ? ParE_toxin ? HNH_2 DndB Uma2 AAA_23 SMC_N 23S_rRNA_IVP PAPS_reduct ?

NZ_NKYE01000032.1:1-4147
Amycolatopsis antarctica

ParE_toxin PhdYeFM_antitox ? ? ? ? ? ? ? ?

NZ_PGGV01000038.1:142871-163156
Glycomyces xiaoerkulensis

? RibD_C ? DUF6247 GATase Chorismate_bind Anth_synt_I_N tRNA-synt_1 Anticodon_1 DUF5915 ParE_toxin HTH_OrfB_IS605 OrfB_IS605 Cas12f1-like_TNB DUF4878 PDDEXK_1 UvrD_C UvrD_C UvrD-helicase DrrA1-3_C ApbA_C ABC_tran PhdYeFM_antitox ? PDDEXK_1 UvrD_C UvrD-helicase Y1_Tnp

NZ_RQGU01000130.1:199301-215127
Leptospira selangorensis

RuvB_N AAA_lid_4 RuvB_C Calci_bind_CcbP Guanylate_cyc HNOBA 7TMR-DISM_7TM TonB_C Thymidylate_kin DUF5982 Omp85 ParE_toxin RelB-like ? NmrA ABC_tran ABC_tran DUF3556 NAD_binding_8 Trypsin PDZ_2 HTH_18

NZ_PCQA01000009.1:47571-67533
Arthrobacter sp. MYb213

DUF3800 TrwC AAA_30 ? ? Aldedh GMC_oxred_C GMC_oxred_N ParE_toxin PhdYeFM_antitox ? ? HTH_7 Resolvase OGG HTH_7 Resolvase SMC_N HTH_18 AraC_binding_2

NZ_QGDQ01000001.1:212566-232249
Quadrisphaera granulorum

Rhodanese ThiF ? MFS_1 ParE_toxin PhdYeFM_antitox AsnC_trans_reg PDDEXK_1 UvrD_C UvrD-helicase MCPsignal HAMP CHASE3 MerR_1 MerR-DNA-bind NUDIX PDDEXK_1 UvrD_C UvrD_C UvrD-helicase CW_binding_2 CW_binding_2 CW_binding_2 ? DNA_binding_1 AAA_16

NZ_QWDC01000001.1:1502127-1518362
Mucilaginibacter conchicola

Peripla_BP_6 Methyltr_RsmB-F MacB_PCD FtsX MacB_PCD FtsX MarC MFS_1 AA_permease_2 RelB-like ParE_toxin ? H_PPase AA_permease_2 ? ? ? Peptidase_S13

NZ_QQXK01000053.1:220-3632
Galactobacter valiniphilus

CSD ? ? ParE_toxin PhdYeFM_antitox RHH_1 FA_desaturase

NZ_RQZF01000007.1:60008-82219
Schaalia canis

RAMPs RAMPs ? ? Pps Pps Pps PhdYeFM_antitox ABC_membrane ABC_tran CbiQ ABC_cobalt GDPD RAMPs ABC_membrane ABC_tran ABC_tran ABC_tran CRISPR_Cas2 ParE_toxin Cas_Cas1

NZ_CAAAFT010000071.1:9208-12925
Candidatus Protofrankia californiensis

PEPCK_N PEPCK_GTP PIN PhoH PMSR PhdYeFM_antitox ParE_toxin Nitroreductase ? ? Acetylase Polyketide_cyc2 NUDIX Beta-lactamase2 HlyIII

NZ_CAKMSC010000002.1:45152-54855
Rothia nasimurium

Ribosomal_L20 Ribosomal_L35p RelB YafQ_toxin PhdYeFM_antitox ParE_toxin DUF1844 SseB His_biosynth ? IF3_C IF3_N Methylase_S Methylase_S N6_Mtase HsdM_N Nitroreductase Nitroreductase

NZ_VDFW01000011.1:22507-33618
Amycolatopsis alkalitolerans

FeS_assembly_P PaaA_PaaC PhdYeFM_antitox ParE_toxin Receiver ANTAR BPD_transp_2 ABC_tran BCA_ABC_TP_C ABC_tran Peripla_BP_6 PaaA_PaaC TetR_N TetR_C_24 PaaB BPD_transp_2 ? Zn_ribbon_PaaD

NZ_BKAU01000001.1:2208899-2217799
Chitinophaga cymbidii

Zn_ribbon_recom Recombinase Resolvase ? SDH_protease DUF2158 ? ParE_toxin ? ? ? ? ? ? HTH_3 DNA_methylase DNA_methylase Vsr

NZ_VOSL01000140.1:18594-38937
Lujinxingia vulgaris

? RecC_C Exonuc_V_gamma PDDEXK_1 UvrD_C UvrD_C UvrD-helicase PhdYeFM_antitox ParE_toxin ? ? HTH_3 HigB-like_toxin HTH_3 Bestrophin-like Trypan_PARP RNase_H_2 UvrD_C_2 SH3_13 AAA_19 CvfB_WH S1_2 S1_2

NZ_JALCMD010000006.1:139182-149179
Bifidobacterium tibiigranuli

PyrI PyrI_C PhdYeFM_antitox ? OMPdecase Amidohydro_1 ? ? HTH_3 ParE_toxin Nitroreductase YcaO OTCace_N OTCace

NZ_AP022810.1:534382-546916
Desulfuromonas sp. AOP6

EamA EamA ? Orn_DAP_Arg_deC Orn_Arg_deC_N AsnC_trans_reg HTH_AsnC-type PhdYeFM_antitox ParE_toxin DUF1847 ? Calcineurin Aminotran_4 Chorismate_bind ABC_tran ABC_tran_Xtn ABC_tran FKBP_C PilZ Receiver ? RUBY_RBDX

NZ_JAAKZI010000001.1:268746-280700
Arthrobacter silviterrae

DUF6278 ? HTH_3 PBP_N EamA EamA Cupin_2 HTH_3 ParE_toxin PhdYeFM_antitox ? ? ? ? GntR Aminotran_1_2 EamA EamA DUF2975

NZ_JAODLR010000018.1:89515-98731
Crystallibacter permensis

? P-II DUF1538 DUF1538 ? Abhydrolase_1 DUF7800 PhoD UPF0158 ParE_toxin Pyridox_ox_2 Pyridox_ox_2 DUF6176 BenE BenE PhdYeFM_antitox

NZ_CP059732.1:482067-496371
Spirosoma foliorum

His-Kinase His-Kinase Apolipoprotein SpoU_methylase bPH_2 MutS_I MutS_II MutS_III MutS_V DUF5677 RelB-like ParE_toxin PAP2 VCPO_N TFR_dimer Peptidase_M28 PA LolA_like Nuc_H_symport DUF6036 Rad52_Rad22 Rad52_Rad22

NZ_CP060716.1:209647-222947
Leucobacter denitrificans

Pep_deformylase Aldo_ket_red ADH_zinc_N ADH_N Lactamase_B DUF488 ParE_toxin PhdYeFM_antitox ICL ICL ICL Peptidase_M1_N Peptidase_M1 ERAP1_C Rv2466c-like LacAB_rpiB HhH-GPD Bac_globin RNase_T MS_channel_1st MS_channel_2nd

NZ_BMIK01000013.1:121744-138137
Parapedobacter defluvii

UDPG_MGDP_dh_N UDPG_MGDP_dh UDPG_MGDP_dh_C Wzz Poly_export SLBB SLBB SLBB_2 SLBB_2 SLBB SLBB HTH_DeoR AAA_14 DUF4143 Rrf2 ParE_toxin RelB-like DUF748 Ldh_2 Peptidase_M1 HupE_UreJ_2 DUF6702 SLFN_AlbA_2 HATPase_c_4 HTH_24 23S_rRNA_IVP

NZ_DASZXR010000018.1:96803-108947
Puia sp.

HTH_11 DUF4292 Cytochrom_C DUF6786 ? HTH_3 RelB-like ParE_toxin Imm33 ? ? ? GFO_IDH_MocA GFO_IDH_MocA_C2 Glucosamine_iso Acetylase

NZ_JACJPM010000079.1:96006-110089
Leptolyngbya sp. FACHB-17

GtrA_DPMS_TM Glycos_transf_2 TPRs TPRs DnaJ tRNA-synt_His GTP_EFTU GTP_EFTU_D2 EFG_III EFG_C BipA_C SQS_PSY ? ParE_toxin AAA_3 AAA_lid_2 FAD_syn Flavokinase Transthyretin DUF559 GatC EVE DUF2281 GHL10

NZ_BAAAMF010000031.1:115550-126435
Pseudarthrobacter sulfonivorans

? NTP_transferase HTH_18 Glyoxalase ABC2_membrane_3 PhdYeFM_antitox ParE_toxin Abhydrolase_1 IclR_C ? HD DJ-1_PfpI HTH_18 GAF_2 ANTAR Cupin_2 HTH_3 ABC_tran DrrA1-3_C

NZ_JACSPR010000001.1:144667-167490
Corynebacterium gallinarum

SRP_SPB SRP54 SRP54_N HD GlnD_UR_UTase NTP_transf_2 P-II Ammonium_transp SRP54 SRP54_N PhdYeFM_antitox ParE_toxin ? SMC_N Acylphosphatase Abhydrolase_6 Na_Ala_symp zf-FPG_IleRS H2TH Fapy_DNA_glyco dsrm Ribonucleas_3_3 MFS_1 MFS_1

NZ_CP063212.1:2304484-2322103
Trueperella pecoris

PhdYeFM_antitox Cas_CT1975 Cas_Cas5d TrmB HATPase_c_4 SLFN_AlbA_2 ? Lipocalin_2 tRNA_U5-meth_tr tRNA_U5-meth_tr ParE_toxin TetR_N TrkA_C TrkA_N TrkH ? ? DUF2871 HrpB_C Helicase_C DEAD

NZ_CP054142.1:1452816-1469100
Treponema parvum

DUF6290 DUF2281 ? PIN ? DUF4474 ? ParE_toxin RHS AMP-binding AMP-binding_C_3 MGDG_synth Glyco_tran_28_C HigB-like_toxin AMP-binding AMP-binding_C_3 ? ThiI THUMP ThiI_fer

NZ_CP071696.1:2828523-2841584
Agromyces archimandritae

Aminotran_1_2 GntR Pyridox_ox_2 EamA EamA MMR_HSR1 YchF-GTPase_C PhdYeFM_antitox ParE_toxin Glyoxalase HTH_3 rve Mu-transpos_C_2 IstB_IS21 ? ? DUF4096 HTH_3 RNase_T Methyltransf_11

NZ_CP076538.1:2001629-2021401
Amycolatopsis aidingensis

CRISPR_Cse2 CRISPR_Cse1 AbiEi_4 ? HTH_31 PhdYeFM_antitox ParE_toxin SH3_3 PhoD_N PhoD PadR Vir_act_alpha_C ABC_tran FtsX FtsX Methyltransf_19 UbiA Cas3_C Cas3-like_C_2 DEAD Cas3_HD

NZ_JAJKFK010000005.1:185036-199329
Amycolatopsis sp. GM8

p450 p450 ? ? ? ? Phage_int_SAM_4 PaaA2 ParE_toxin HAD_2 GntR HTH_17 Sigma70_r2 Sigma70_r4_2 HTH_17 AAA_25 HTH_17 Phage_int_SAM_3 Phage_integrase Phage_integrase

NZ_JAMXEX010000004.1:141947-161344
Ancylothrix sp. D3o

? TPRs TPR_CcmH_CycH TPRs TPRs TPRs Guanylate_cyc CHASE2 RPE65 ? Met_synt_B12 B12-binding B12-binding_2 Pterin_bind S-methyl_trans PhdYeFM_antitox ParE_toxin ? DUF488 Toxin_YhaV PrlF_antitoxin PAS PAS PAS His-Kinase His-Kinase DAGK_cat YegS_C CHASE3 PAS PAS PAS PAS His-Kinase His-Kinase

NZ_CP113264.1:2745639-2762164
Streptomonospora nanhaiensis

DUF1028 Peptidase_M20 Lyase_1 FumaraseC_C MFS_1 MFS_1 adh_short HTH_31 MLTR_LBD ParE_toxin ? GTP_cyclohydroI QueC GerE TPRs AAA_16 Methyltransf_11 MarR_2 PhdYeFM_antitox Radical_SAM

NZ_CAMEMO010000030.1:9282-19319
Treponema succinifaciens

SSB ? ? ? ? BrnA_antitoxin ParE_toxin DUF6290 Toprim Topoisom_bac Toprim_Crpt ? ? DNA_methylase DpnD-PcfM Phage_CI_repr BrnT_toxin

NZ_JAQKQJ020000006.1:91758-104957
Nodosilinea sp. PGN35

Mem_trans Mem_trans HicB_lk_antitox Uma2 Aldedh DUF29 ThiC_Rad_SAM ParE_toxin ? Glyoxalase_6 Fic_N Fic ? DUF4383 SSF DUF3782 DUF29

NZ_CP112874.1:4468736-4483039
Pseudanabaena galeata CCNP1313

? His-Kinase His-Kinase PAS HAMP Uma2 Yae1_N ? PhdYeFM_antitox BCA_ABC_TP_C ABC_tran ParE_toxin ? DUF3326 MFS_1 MFS_1 Pps Pps Pps Pps Pps TerC FAD_binding_3 Glyoxalase

NZ_WHTD01000148.1:11679-20962
Actinophytocola sp.

CorC_HlyC CBS CBS CNNM CNNM ? SecDF_P1_head PCMT ParE_toxin ? ? ? DUF6879 MDMPI_N Transglut_core3 Transglut_core3

NZ_CABUED010000070.1:2306-9769
uncultured Fusobacterium sp.

? MazE_antitoxin ? ? YqaJ RecT ? ? DUF6290 ParE_toxin Resolvase Recombinase ? Phage_pRha ?

NZ_JAJOIL010000028.1:154820-169167
Seleniivibrio sp.

Sacchrp_dh_C Sacchrp_dh_NADP Arg_decarbox_C Arg_decarb_HB Orn_Arg_deC_N His-Kinase His-Kinase Receiver NtrC-AAA AAA_lid_14 HTH_8 DUF2860 Pyr_redox_2 NDH2_C RHH_3 ParE_toxin ? ? BamE ? AAA_25 Phage_AlpA bPH_2

NZ_JAQSCS010000017.1:31571-50454
Methylobacter sp.

DUF4912 GH94_b-supersand AAA_15 AAA_15 OLD-like_TOPRIM Unstab_antitox ? RelB-like ParE_toxin Y1_Tnp Glyco_hydro_30C Glyco_hydro_30 ISOA1-3_C Alpha-amylase CBM_48 ChlI AAA_25 Zn_ribbon_LapB Zn_protease CorA Peptidase_M28 BE_C Glyco_hydro_57

NZ_CP129884.1:439764-453396
Methylomonas montana

Cas_Cas5d CRISPR_assoc MazE_antitoxin PIN HTH_37 DUF6290 ParE_toxin Cas_Cas1 Cas_Cas1 Cas_Cas2CT1978 Fer4_21 FeS tRNA_bind Anticodon_3 tRNA-synt_1g ? ? SPOR Ldt_C YkuD LysM CRISPR_assoc

NZ_CAMCLB010000026.1:25112-37381
Treponema berlinense

? NLPC_P60 GGDEF META ATP_bind_3 Lactamase_B Beta-Casp RMMBL DUF6290 ParE_toxin Big Pectate_lyase_4 AAA-ATPase_like

NZ_CAUVXG010000075.1:1375-13258
Frankia sp. Cr2

adh_short_C2 ? ? RHS RHS_repeat RHS_repeat RHS_repeat RHS_repeat WXG100_2 ? ParE_toxin AhpC-TSA Methyltransf_25 Thioredoxin_4 MauE SCO1-SenC ? DUF3618

NZ_JAYGHW010000011.1:22007-38933
Spirulina sp. CCNP1310

DHBP_synthase GTP_cyclohydro2 EVE ? ? ? ? ParE_toxin ? ? PemK_toxin ? RNaseH-CATRA+Caspase TPRs TPRs TPRs TPRs TPRs TPRs PPC Trypsin MutS_V MutS_III MutS_II MutS_I Fer4_5 cNMPBD

NZ_JAYFUM010000014.1:143393-155569
Arcicella rigui

PIN Arginosuc_synth Arginosuc_syn_C DUF2281 PIN Semialdhyde_dh Semialdhyde_dhC_1 ? ? ParE_toxin Aminotran_3 Peptidase_S9 DPPIV_N ? LuxE MoaE DHFR_1 Formyl_trans_N ACT

NZ_JAYGHP010000008.1:12041-21083
Crocosphaera sp. UHCC 0190

Semialdhyde_dh Semialdhyde_dhC_1 NTP_transf_9 Glyoxalase DUF697 MMR_HSR1 LMBR1 ? PseudoU_synth_2 ParE_toxin ? HTH_3 ? DUF6883 DUF4926 HicB_lk_antitox Y1_Tnp Uma2

NC_031072.1:24924-32659
Gordonia phage CaptainKirk2

? fn3 SASA ? ? ? Peptidase_C27 PhdYeFM_antitox ParE_toxin Trypan_PARP HTH_31 ? Phage_integrase ? HTH_31 ?